



Federal Office
for Information Security

5G Risk Analysis

Detailed Analysis: **Transport Network**



Version History

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1 List of Abbreviations

ADM	Add-Drop Multiplexer
BGP	Border Gateway Protocol
BH	Backhaul
BVI	Bridged Virtual Interface
CES	Carrier Ethernet Switch
CP	Control Plane
DoS	Denial of Service
EIN	Evolved IP Networks
ENM	Ericsson Network Management
EoC	Ethernet over Copper
EoF	Ethernet over Fiber
FH	Fronthaul
GMPLS	Generalized Multi-Protocol Label Switching
GTP	GPRS Tunneling Protocol
IP	Internet Protocol
IP RAN	IP Radio Access Network
IPv6	Internet Protocol Version 6
KPI	Key Performance Indicator
L3VPN	Layer 3 Virtual Private Network
LDP	Label Distribution Protocol
LLVD	Load Low Voltage Disconnect
MBH	Mobile Backhaul
MNO	Mobile Network Operator
MO	Managed Object
MOM	Managed Object Model
MPBN	Mobile Packet Backbone Network
MPLS	Multi-protocol Label Switching
MeNB	Master eNodeB
NOC	Network Operation Center
NSA	Non-Standalone
OAM	Operation, Administration, Maintenance
OLT	Optical Line Terminals
ONU	Optical Network Units
OSPF	Open Shortest Path First
OTN	Optical Transport Network
PIM	Protocol Independent Multicast
PON	Passive Optical Network
PTP	Precision Time Protocol
PWE3	Pseudowire Emulation Edge-to-Edge
RP	Router Processor
RSVP-TE	Resource Reservation Protocol Traffic Engineering
SA	Standalone
SCTP	Stream Control Transmission Protocol
SDN	Software Defined Networking
SSL	Secure Sockets Layer
SyncE	Synchronous Ethernet
TCPIP	Transmission Control Protocol over IP

TLS	Transport Layer Security
TN	Transport Network
UDP	User Datagram Protocol
UP	User Plane
UPS	Uninterruptable Power Supply
VPWS	Virtual Private Wire Service
X2AP	X2 Application Protocol

2 Introduction and Purpose

2.1 Intention

The intention of the document is to dive deep into the analysis of risk scenarios where the **transport network** is the attacked or failing subsystem of a mobile network.

This document shows the current view on the probability that any risk scenario could become reality.

The study is only available in English language, as the target group is primarily the international IT security community. The study is also in the context of English publications on this topic.

2.2 Link to Framework Document

This document is linked to the framework document and provides relevant information about transport network technology and a probability assessment of relevant risk scenarios described in the framework document. The overall risk assessment for each risk scenario consists of the results of this analysis and the impact described in the framework document. The overall risk assessment for the transport network is summarized in Chapter 6 of this document and in the framework document.

3 Methodology

The methodology used for the 5G Risk Analysis (5G-RA) is based on PASTA. In this document and the related documents, we will refer to the methodology as 5G-RA.

This chapter explains the seven steps of the 5G-RA methodology and assigns them to the corresponding PASTA steps.

Step 1: Definition of Risk Scenarios

In the first step of the 5G-TARA, the 5G network needs to be analysed from the following three perspectives:

1. Public Mobile Network Operator (MNO) Perspective,
2. Private 5G Network Operator Perspective,
3. User Perspective.

Several risks with a significant impact on the society or public safety have been identified by the legislative in the past. To mitigate these risks, laws were set into force, e.g. the legislative to ensure emergency calls are available (TKG §165). Most of these laws define fines for mobile network operators if the regulative obligation is not followed.

These perspectives are explained in detail in Chapter 6 of the framework document. The goal is to identify the possible risks that the 5G network might be exposed to from each perspective.

Step 2: 5G System Diagram

In step 2, a detailed system diagram of the parts of a 5G network needs to be created. This diagram will serve as a basis for the following steps.

Step 3: Refinement of Detailed 5G Subsystem Diagrams

In this step, more technical definitions and explanations of the relevant 5G subsystems are introduced. For example, various network functions are part of the 5G core network. This level of detail serves as a basis for the following steps.

Step 4: Collection of Known 5G Threats

In this step, publicly known 5G specific threats are collected. For this purpose, relevant studies and input documents are collected. This is the basis for the identification of threats in the next step.

Step 5: Search of 5G Vulnerabilities

Step 5 includes the analysis and identification of relevant threats for each risk scenario. The findings of this step are part of the corresponding Detailed Risk Analyses.

Step 6: Examination of Identified Threats

In this step, all threats that were found in step 5 are examined on whether there is a possibility for this threat to be realized. Hypothetical scenarios are used as well as references to successful attacks, which were performed in the past.

Step 7: Risk Calculation

The last step of the methodology is a calculation of the risk for all threats that were identified in step 5 and are still considered relevant after step 6 was carried out. The overall results are presented in Chapter 6 of this document.

The following table assigns the steps of the 5G-RA to the steps of PASTA and gives an overview of the document containing the results of each step.

Table 1: Assignment of 5G-RA steps to PASTA-steps

#	5G-RA methodology	Document Layer	PASTA
1	Definition of Risk Scenarios	Framework	Define Objectives
2	5G System Diagram	Framework	Define Technical Scope
3	Refinement of Detailed 5G Subsystem Diagrams	Framework & Detailed Risk Analysis	Application Decomposition
4	Collection of Known 5G Threats	Detailed Risk Analysis	Threat Analysis
5	Search of 5G Vulnerabilities	Detailed Risk Analysis	Vulnerability and Weakness Analysis
6	Examination of the Identified Threats	Detailed Risk Analysis	Attack Modelling
7	Risk Calculation	Framework & Detailed Risk Analysis	Risk and Impact Analysis

4 5G Network Model: Transport Network

The 5G network model introduced in the framework document is used to split the 5G risk analysis into multiple Detailed Risk Analyses. The network model follows standards (like 3GPP standards) as much as possible and comprises a variety of interconnected subsystems that are categorized on a high-level view as shown in Figure 1. 5G networks may consist of any combination of 5G Standalone and 5G Non-Standalone architecture options, which therefore includes also 4G RAN (E-UTRAN) and 4G-CORE (EPC). However, this risk analysis covers only 5GC for the core. All other subsystems are not limited to 5G.

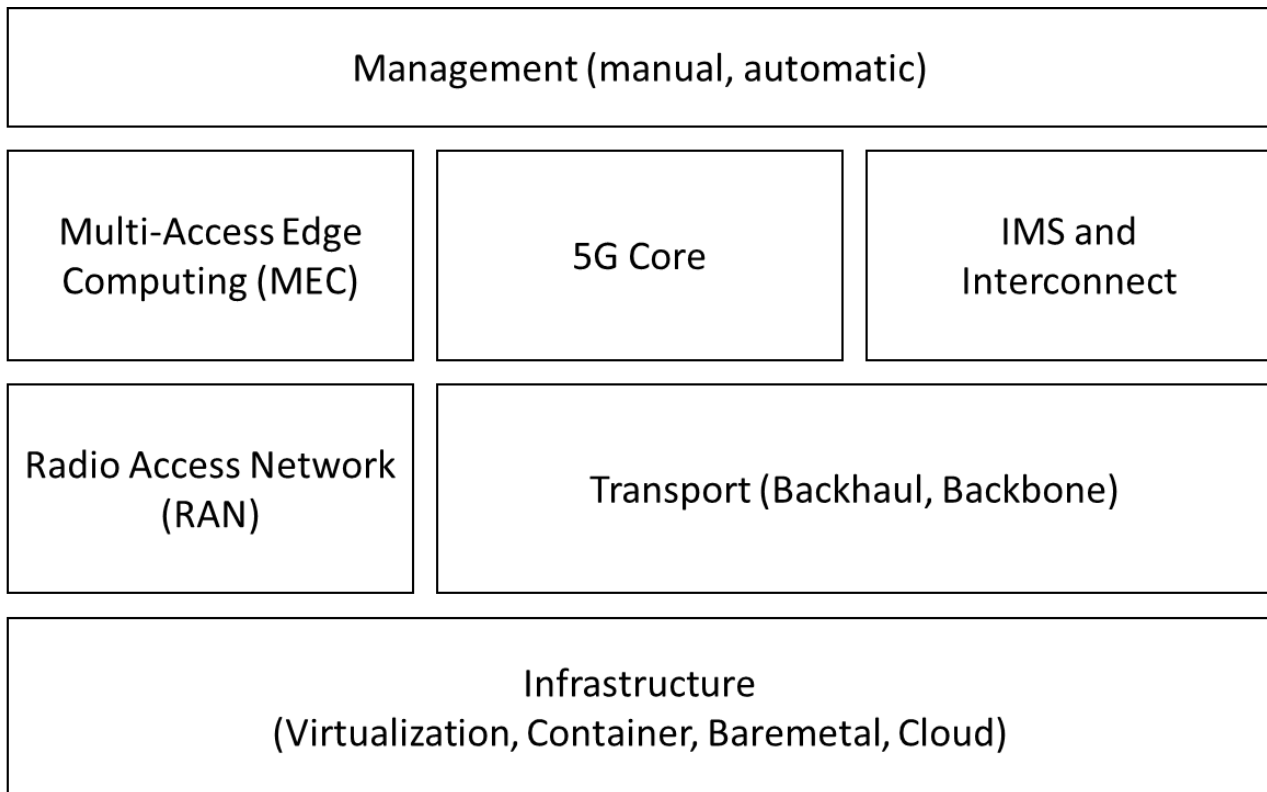


Figure 1: 5G Network Model: High level view

4.1 Definitions

Term	Definition
NOC	Network Operation Center, in which the network can be monitored and controlled.
Optimizer	Role that monitors the key performance indicators (KPI), finds solutions to improve the KPI and implements them in the network.
Pilot	Small-scale test network used to evaluate and validate new technologies or protocols before wider implementation.
Service affect	Activities which effect the services on the network.
Upgrade package	A software package which is provided by the equipment vendor.

4.2 Introduction to Transport Networks

4.2.1 Transport Infrastructures

Telecommunication infrastructure encompasses a wide range of technologies and facilities for transmitting data and delivering communication services.

Different components of telecommunication infrastructure are:

Component	Description
Backbone Networks	High-capacity network of that interconnects data centers. It carries aggregated traffic from backhaul networks and provides connectivity for long-haul and inter-city traffic.
Cable Landing Stations	Facilities where undersea cables are terminated on land.
Cell Towers	Cellular network infrastructure that provides wireless communication coverage.
Cloud-Computing Infrastructure	Data storage and processing resources provided over the internet.
Data Centers	Facilities for storing and processing vast amounts of digital data and hosting web services.
Data Transmission Protocols	Standards and protocols governing how data is transmitted and received, such as TCP/IP.
Fiber Optic Cables	High-capacity, high-speed data transmission through thin strands of glass or plastic fibers.
Fiber Optic Distribution Hubs	Facilities for splitting and distributing fiber optic connections to users.
Fiber Rings	High-capacity fiber optic networks that encircle cities or regions.
Aggregation Points	Aggregation points are locations within a network where data from multiple sources or devices is collected, aggregated, and then forwarded to its destination.
Internet Exchange Points (IXPs)	Locations where internet service providers (ISPs) exchange traffic.
Last-Mile Infrastructure	The final connection between a service provider and the user, often through DSL, cable, or fiber.
Microwave Towers	Used for point-to-point data transmission, often in remote areas.
Routers and Network Equipment	Hardware for routing and managing data traffic within networks.
Satellite Ground Stations	Facilities for receiving and transmitting signals to and from satellites in orbit.
Switching Centers	These manage the routing of voice and data communications within a network.

Component	Description
Telecommunication Satellites	Orbiting satellites used for television broadcasting, GPS, and long-distance communication.
Telecom Software	Systems and software for managing network operations, billing, and security.
Telecom Towers	Structures that support antennas for wireless communication.
Undersea Cables	Submarine cables that connect continents for international data transmission.
Wireless Access Points (WAPs)	Devices that enable wireless connectivity in areas like Wi-Fi hotspots.

These components work together to enable the transmission of voice, data and multimedia content over short and long distances, supporting various forms of communication, including telephone, internet, television, and more.

4.2.2 Comparison of Fiber Optic and Microwave Networks

Fiber optic and microwave links are two common and important technologies used in telecommunication networks for data transmission.

In the following we delve into a more detailed comparison of fiber optic and microwave links in telecommunication networks, including some key figures:

Key Parameter	Fiber Optic	Microwave
Data Capacity	<i>Very High</i> Fiber optic cables have an incredibly high data capacity, measured in terabits per second (Tbps) or more. For example, a single strand of modern fiber can transmit up to 25 Tbps or more.	<i>Medium</i> Microwave links have a lower data capacity compared to fiber optics, typically ranging from 155 Mbps to 10 Gbps per link, depending on the frequency band used.
Latency	<i>Low</i> Fiber optic links have extremely low latency, typically in the range of 1 to 10 milliseconds (ms).	<i>Low</i> Microwave links also offer low latency, ranging from 1 to 5 ms, but the propagation can be slightly faster than fiber.
Distance	<i>Long-Haul</i> Fiber optic cables can transmit data over very long distances without significant signal loss. For example, undersea fiber cables can span thousands of kilometers.	<i>Short to Medium in Urban Area / Long-Haul</i> Microwave links are generally used for shorter to medium distances, typically up to 50 kilometers in point-to-point configurations.
Bandwidth	<i>Up to 1.84 Petabit (10^{15}) per second</i> Fiber optics can provide nearly unlimited bandwidth, making them suitable for high-demand applications like streaming, cloud computing, and data centers.	<i>Limited to 10 Gigabit (10^9) per second</i> Microwave links have limited bandwidth compared to fiber, which can be a bottleneck for data-intensive applications.
Security	<i>Highly Secure</i> Fiber optic communication is highly secure because it is difficult to intercept signals without physically tapping the cable.	<i>Requires Encryption</i> Microwave signals can be intercepted with the right equipment if not properly secured.

Key Parameter	Fiber Optic	Microwave
Reliability	<i>High</i> Fiber optics are highly reliable with minimal maintenance required once installed.	<i>Good</i> Microwave links require periodic maintenance and are susceptible to atmospheric conditions that can affect signal quality. Microwave links require a clear line of sight between antennas. Obstacles like buildings or terrain can disrupt the signal.
Deployment Time	<i>Longer</i> Deploying fiber optic networks can be time-consuming, involving digging and laying cables.	<i>Shorter</i> Microwave links can be deployed relatively quickly, often within days, making them suitable for rapid network expansion.
Cost	<i>High (Initial)</i> Initial installation costs for fiber optics can be high, especially for long-distance projects.	<i>Lower (Initial)</i> Microwave links can be cost-effective for shorter distances, but the cost may increase for longer links or in challenging terrain.

In summary, the choice between fiber optic and microwave links in telecommunication networks depends on various factors, including the required data capacity, distance, latency sensitivity, and cost considerations. Fiber optics excel in providing high-speed, high-capacity, and secure long-distance connectivity, while microwave links are more suitable for shorter-distance connections, rapid deployment, and situations where line of sight can be maintained. Often, a combination of both technologies is used in telecommunication networks to optimize performance and cost-effectiveness.

However, long haul microwave solutions are used where there is a need for high capacities over relative long distances in rural and remote areas. Due to the cost efficiency of microwave products it is also increasingly used in areas where optical fiber technically can be an option, but the deployment of fiber would be more expensive.

Microwave long haul systems utilize the frequency range of 5 – 13 GHz. They provide the best-in-class spectrum efficiency and make the most out of the available frequency spectrum. Typical for long haul systems is that they use many channels over a hop. They can enable capacities up to 10 Gbps per hop, and make very long hops, up to 200 km. The long-haul systems can enable capacities up to 10 Gbps per hop over up to 35 km. It can also make very long hops, up to 200 km long with a lower capacity. The regular use cases for long haul are creating a backbone network in remote or rural areas, redundancy for fiber and reaching sites where fiber is difficult or too costly, like over sea or across a deep valley.

4.3 Backhaul and Backbone

Backhaul and backbone networks work in tandem to provide end-to-end connectivity in a telecommunications transport network. The backhaul connects users to the core network, while the backbone handles long-distance and interconnection traffic. Both are important for ensuring reliable and high-performance telecommunications services. Both types of networks are described in more detail in the next sections. [16]

4.3.1 Traditional Backhaul

Backhaul refers to the network segment that connects the MNO's local access network (e.g., cell towers, DSLAMs, cable headends) to the core or backbone network. [23]

Components and Technologies	Description
Access Points	These are the points of connection where traffic from users is aggregated, such as cell towers or customer premises.
Fiber and Microwave Links	Backhaul networks typically use high-capacity links, often based on fiber optics, or microwave links for remote or difficult-to-reach areas.
Access Aggregation Equipment	Devices like high-capacity switches and routers are used to aggregate traffic from multiple access points.
Edge Routers/Switches	These devices route data within the backhaul network. They determine how data should be forwarded from the aggregation points to the core network, considering factors like network conditions and quality of service requirements.
Quality of Service (QoS)	Backhaul networks prioritize traffic based on different service requirements, ensuring low latency and high bandwidth for important applications like voice and video.
Protocols	Backhaul networks use various protocols, including Ethernet, IP/MPLS (Multi-Protocol Label Switching), and SONET/SDH (Synchronous Optical Networking/Synchronous Digital Hierarchy) for resilience.

Key Considerations:

- **Capacity Planning:** Backhaul networks must be dimensioned to handle the aggregated traffic from access points.
- **Latency:** Low latency is crucial for real-time applications like VoIP and video streaming.
- **Redundancy:** This means having backup connections and routes in case of failures, ensuring uninterrupted service.
- **Security:** Encryption and security measures protect the confidentiality and integrity of the transported data.

4.3.1.1 5G Backhaul

Definition

The 5G backhaul is a crucial segment of the 5G network that connects 5G cell sites (such as small cells, macro cells, or base stations) to the central 5G core network. It consists of high-capacity, low-latency links designed to transport data traffic generated by connected devices from the network's edge to its core. It also refers to the RAN-Core network interface (N1, N2, N3).

Function

The primary role of the 5G backhaul is to aggregate and transport data traffic generated by 5G-connected devices to the 5G core network efficiently. It ensures that data, including voice, video, and internet traffic, can seamlessly flow from the network's edge (cell sites) to the core, where it can be routed to its final destination.

Characteristics

High-capacity links are typically employed to handle the substantial increase in data traffic that 5G networks are expected to generate. Backhaul technologies can include fiber optic connections, wireless links (e.g., millimeter-wave wireless), and, in some cases, satellite connections.

The backhaul network must be designed for reliability and low latency to support applications such as real-time video streaming and IoT device communication.

In summary, the 5G backbone and 5G backhaul are integral components of a 5G network infrastructure. The 5G backbone is responsible for routing data within and across core networks, while the 5G backhaul focuses on connecting 5G cell sites to the core network, ensuring efficient and low-latency data transport, and together they enable the advanced capabilities of 5G technology.

4.3.1.2 Backhaul Equipment

Components and Technologies	Description
Fiber-Optic Links	Fiber-optic cables are the primary choice for high-capacity backhaul in 5G networks. They offer low latency, high bandwidth, and excellent reliability. Backhaul routers and switches are used to manage and route traffic over these fiber-optic links.
Microwave Links	Microwave links are used for wireless backhaul in areas where laying fiber-optic cables is challenging or cost-prohibitive. These links use radio waves to transmit data between cell sites and the core network.
Ethernet Backhaul	Ethernet-based backhaul solutions, including Carrier Ethernet switches and routers, provide high-speed, cost-effective connectivity between base stations and the core network.
GPON (Gigabit Passive Optical Network)	GPON technology can be used in areas where fiber-optic backhaul is not readily available. It provides high-speed broadband connectivity over existing passive optical networks.

4.3.2 5G IAB (Integrated Access Backhaul)

5G IAB stands for "5G Integrated Access and Backhaul". It is a technology that allows for the integration of access and backhaul functions in 5G wireless networks. In traditional cellular networks, access (connecting user devices to the network) and backhaul (connecting cell sites to the core network) are separate elements. With 5G IAB, these functions can be combined, simplifying network architecture and potentially improving network efficiency, especially in situations where it is challenging to deploy traditional backhaul infrastructure. This can be particularly useful in scenarios like dense urban environments or remote areas.

The function of 5G IAB, or Integrated Access and Backhaul, involves the integration of access and backhaul capabilities within a single network node or device. Here is a more detailed explanation of its functions:

- **Access:** 5G IAB provides wireless access to user devices, such as smartphones, IoT devices, and other connected devices. It uses advanced 5G radio technology to connect user devices to the network, offering high data rates and low latency.
- **Backhaul:** Backhaul is the part of the network that connects cell sites (base stations) to the core network infrastructure. It typically involves high-capacity links to transport large amounts of data. In traditional networks, backhaul is a separate and often wired network. However, 5G IAB combines access and backhaul functions, reducing the need for a separate backhaul network.
- **Integration:** 5G IAB integrates both access and backhaul into a single network element, such as a cell site or base station. This integration simplifies network deployment and management, making it more cost-effective and efficient.
- **Flexibility:** 5G IAB can be particularly useful in scenarios where it is challenging to deploy traditional wired backhaul infrastructure, such as in rural or remote areas. It allows for more flexible deployment options, as it can use wireless links for backhaul connections. IAB can be scaled more easily to meet

the demands of different deployment scenarios. This is particularly important in scenarios where rapid network expansion is required.

- **Efficiency:** By integrating access and backhaul, 5G IAB reduces the need for duplicate hardware and infrastructure, potentially making the network more efficient and cost-effective.
- **Standardization:** IAB technology is developed and standardized by organizations like the 3rd Generation Partnership Project (3GPP), ensuring interoperability and compatibility across different network equipment and vendors.

Overall, 5G IAB enhances the 5G network's flexibility and efficiency by combining the functions of providing wireless access to users and connecting cell sites to the core network, all within a single integrated system. This technology is part of the ongoing evolution of cellular networks to meet the growing demands for high-speed, low-latency connectivity.

4.3.3 Backbone

The backbone network is the high-capacity subsystem of telecommunication networks that interconnects various regions, cities, and data centers. It carries aggregated traffic from backhaul networks and provides connectivity for long-haul and inter-city traffic.

Components and Technologies:

- **Key Routers and Switches:** High-capacity routers and switches form the core of the backbone network.
- **DWDM (Dense Wavelength Division Multiplexing):** Optical technology that enables multiple data streams to be sent over a single fiber optic cable using different wavelengths.
- **MPLS (Multi-Protocol Label Switching):** MPLS is used to efficiently route and manage traffic within the backbone network.
- **Peering Points:** Interconnection points with other service providers and internet exchange points.
- **Protocols:** The backbone network primarily uses IP/MPLS for routing and managing traffic. BGP (Border Gateway Protocol) is often used for inter-domain routing.

Key Considerations:

- **Scalability:** Backbone networks must be designed to accommodate increasing data volumes.
- **Redundancy and Resilience:** Redundant links and equipment, along with fail over mechanisms, are essential for high availability.
- **Traffic Engineering:** Optimization of traffic flows to ensure efficient utilization of network resources.
- **Quality of Service (QoS):** Prioritization of traffic for different services to ensure performance and reliability.

4.3.3.1 5G Backbone

Definition

The 5G backbone is the central, high-capacity and low-latency core network infrastructure. It serves as the primary communication pathway that connects various network elements, data centers, and points of presence (PoPs) within the 5G network architecture.

Function

The 5G backbone facilitates the rapid and efficient transmission of data between different parts of the 5G network, including connecting cell sites (base stations) to essential network elements. It plays a pivotal role in managing network traffic, routing data between cell sites, data centers, and external networks (e.g., the

internet). The backbone network is responsible for ensuring low-latency communication, high data rates, and reliable connectivity in the 5G ecosystem.

Characteristics

High-speed and high-capacity connections, often utilizing fiber optic cables and advanced routing and switching technologies. Low latency to support real-time applications like augmented reality (AR), virtual reality (VR), and autonomous vehicles.

4.3.3.2 Backbone Equipment

- **Key Routers and Switches:** Routers and switches are fundamental networking devices in the backbone network. Routers are responsible for forwarding data packets between networks, making routing decisions based on IP addresses. Switches operate at the data-link layer (Layer 2) and are used to connect devices within the same network segment.
- **DWDM (Dense Wavelength Division Multiplexing):** DWDM technology allows multiple data streams to be transmitted simultaneously over a single optical fiber by using different wavelengths of light. It is used in the backbone network to increase capacity and reduce latency.
- **Edge Routers:** Edge routers are strategically placed at the edge of the core network to provide efficient routing and access control. They help in managing traffic coming from various access points, including cell sites.
- **Optical Transport Network (OTN) Equipment:** OTN equipment is used to transport data across optical networks efficiently. It supports features like grooming and traffic management in the backbone network.
- **Network Aggregation Equipment:** This includes aggregation routers and switches that consolidate traffic from multiple cell sites or access points before routing it further into the core network.
- **Security Appliances:** Security appliances and firewalls are essential in the backbone network to protect against cyber threats and unauthorized access.
- **Load Balancers:** Load balancers help distribute network traffic evenly across multiple servers or links, ensuring optimal performance and redundancy in the backbone network.
- **Redundancy and Fail-over Equipment:** To ensure network reliability, redundant equipment and fail-over mechanisms are deployed to automatically switch to backup paths or devices in case of failures.

4.3.4 Risks and Threats of Backbone and Backhaul

The transport of backbone and backhaul networks is important for the overall connectivity and performance of telecommunication systems. Risks and threats associated with these networks include:

- **Physical Infrastructure Vulnerabilities:** These networks consist of a complex web of physical components like fiber optic cables, microwave links, and transmission equipment. Physical damage due to natural disasters, accidents, or vandalism can disrupt services.
- **Cybersecurity Threats:** Backbone and backhaul networks are susceptible to cyberattacks, including DDoS attacks, data breaches, and network intrusion attempts. Securing these networks against cyber threats is crucial.
- **Unauthorized Access:** Unauthorized access to important infrastructure points, such as data centers, network nodes, or transmission facilities, can lead to data theft or network disruption.
- **Traffic Congestion:** Congestion can occur when network traffic exceeds network capacity. This can lead to network slowdowns or outages, impacting the quality of service.
- **Spectrum Interference:** Interference from other wireless transmissions, especially in backhaul networks that use microwave links, can degrade signal quality and disrupt communications.
- **Equipment Failure:** Equipment failures in the backbone and backhaul network can disrupt connectivity. Regular maintenance and redundancy planning are essential to mitigate this risk.

- **Supply Chain Risks:** The sourcing and integration of network equipment pose supply chain risks, including the potential for compromised or malicious components.
- **Geopolitical Risks:** These networks often cross-national borders, making them vulnerable to geopolitical disputes or restrictions that can disrupt international connectivity.

Additionally, the fusion of Access and Backhaul in 5G Integrated Access and Backhaul (IAB) brings further potential threats. Here are additional threats specific to 5G IAB:

- **Access-Backhaul Crosstalk:** The integrated nature of IAB poses a risk of unintended crosstalk between access and backhaul functions. This could result in interference, performance degradation, or unauthorized access to backhaul resources.
- **Dynamic Resource Allocation Challenges:** The dynamic resource allocation in IAB introduces risks related to misconfigurations or vulnerabilities in allocation algorithms. This may lead to suboptimal performance, network congestion, or service disruptions.
- **Synchronization Issues:** Synchronization is pivotal in IAB for coordinated access and backhaul operation. Risks include synchronization failures due to misconfigurations or clock inaccuracies, leading to performance degradation or connectivity loss.
- **Backhaul Link Saturation:** Integration of access and backhaul may lead to scenarios where a surge in access traffic overwhelms backhaul links, causing bottlenecks, delays, or temporary backhaul connectivity loss.
- **Access-Backhaul Security Boundary:** Establishing and maintaining a clear security boundary between access and backhaul functions is crucial. Failure to do so may result in unauthorized access to main backhaul infrastructure, leading to potential security breaches.
- **Dynamic Network Slicing Challenges:** IAB supporting network slicing introduces risks like misconfigurations or vulnerabilities impacting service quality or allowing unauthorized access to specific slices.
- **Interference in Multihop Backhaul:** In multihop backhaul scenarios involving relay nodes, risks include interference, signal degradation, or vulnerabilities in relay nodes affecting signal integrity.
- **Backhaul Routing Vulnerabilities:** Risks associated with backhaul traffic routing, such as misconfigurations or vulnerabilities in routing protocols, may cause suboptimal routing or potential security breaches.
- **User Plane Security Challenges:** Ensuring user plane traffic security in IAB is crucial. Risks include vulnerabilities in user plane protocols leading to unauthorized access or data interception.
- **Service Continuity Challenges:** Failures or misconfigurations in the integrated structure may pose challenges to service continuity during handovers or dynamic network changes.

For organizations deploying 5G IAB, conducting thorough risk assessments tailored to the integrated structure's unique characteristics is crucial. Implementing robust security measures, regular audits, and staying informed about emerging threats are vital for effective risk mitigation.

4.4 Protocols used in Transport Networks

The following protocols are typically used in transport networks. Each of these protocols might have their own associated risks. However, this document cannot explain all of the protocols in detail. [20], [32]

Protocol	Abbreviation
Ethernet over Fiber	EoF
Ethernet over Copper	EoC
Internet Protocol Version 6	IPv6
Multiprotocol Label Switching	MPLS
GPRS Tunneling Protocol	GTP
User Datagram Protocol	UDP
Transmission Control Protocol over IP	TCPIP
X2 Application Protocol	X2AP
Stream Control Transmission Protocol	SCTP
Internet Protocol	IP
Synchronous Ethernet	SyncE
Precision Time Protocol	PTP
Border Gateway Protocol	BGP
Transport Layer Security	TLS
Secure Sockets Layer	SSL
Protocol Independent Multicast	PIM
Resource Reservation Protocol - Traffic Engineering	RSVP-TE
Generalized Multiprotocol Label Switching	GMPLS
Label Distribution Protocol	LDP
Pseudowire Emulation Edge-to-Edge	PWE3

4.5 Types of Components used in Transport Networks

- Add-Drop Multiplexer (ADM)
- OTN (Optical Transport Network) Switches
- Optical Amplifiers
- Optical Cross-Connects (OXC)
- Packet Brokers

4.5.1 Add-Drop Multiplexer (ADM)

ADM stands for "Add-Drop Multiplexer." An Add-Drop Multiplexer is a crucial component used to manage the flow of data in optical fiber networks. While ADMs are more commonly associated with traditional optical networks, they still play a role in certain aspects of 5G transport networks, particularly in the backhaul and fronthaul portions.

ADMs are used in a 5G transport network as follows:

- **Optical Network Backbone:** In the backbone of a 5G transport network, ADMs are used to multiplex and demultiplex multiple optical signals onto a single optical fiber. They can add (insert) and drop (extract) individual wavelengths (channels) from the optical signal without disrupting the overall flow of data. This flexibility allows network operators to efficiently manage the allocation of bandwidth for various services and routes.
- **Wavelength Division Multiplexing (WDM):** ADMs are often integrated with Wavelength Division Multiplexing technology, which enables multiple data streams to be transmitted simultaneously over different wavelengths of light on the same fiber. WDM and ADMs help increase the capacity and efficiency of optical networks, supporting the high bandwidth demands of 5G.

- **Ring Topologies:** In some 5G transport networks, especially those with ring topologies for redundancy and reliability, ADMs play an important role in ensuring that data traffic can be efficiently added and dropped at various points in the ring. This helps maintain connectivity even if a portion of the network experiences a failure.
- **Fronthaul and Backhaul:** In the context of 5G fronthaul and backhaul networks, ADMs are used to manage the optical connections between remote radio heads (RRHs) or small cells and the central aggregation points. They allow for efficient aggregation and distribution of data traffic from numerous cell sites to the core network.
- **Network Management:** ADMs are typically managed using network management systems. Network operators can use these systems to monitor and control the optical connections, allocate bandwidth as needed, and perform remote configuration and troubleshooting.

While 5G transport networks often rely on advanced technologies like Ethernet and IP for packet-based data transport, optical technologies and ADMs continue to play a vital role, especially for the high-capacity and low-latency requirements of 5G. ADMs help optimize the use of optical fiber resources and contribute to the overall performance and reliability of the network.

4.5.2 OTN (Optical Transport Network) Switches

The Optical Transport Network (OTN) holds significant importance within 5G transport networks, particularly in the backhaul and core network segments. Here's how OTN functions in a 5G transport network:

- **High-Capacity Transport:** OTN serves as a standardized and high-capacity transport infrastructure, efficiently handling a substantial volume of 5G traffic across diverse network segments. OTN is defined in various ITU Recommendations. Its robust design allows seamless transportation of data between cell towers, data centers, and essential network elements.
- **Packet and TDM Integration:** OTN offers support for both packet and Time-Division Multiplexing (TDM) traffic. This flexibility is pivotal in 5G networks, catering to the diverse mix of traffic types. TDM supports legacy services, ensuring compatibility, while packet-based traffic accommodates the escalating data demands inherent in 5G networks.
- **Traffic Aggregation:** OTN equipment plays a crucial role in aggregating traffic originating from multiple cell sites. By consolidating data streams from various locations, OTN reduces network management complexities and diminishes the requirement for a multitude of leased lines. This aggregation capability streamlines network operations while efficiently managing and transporting traffic across the 5G network.

Through these functionalities, OTN contributes significantly to the efficacy and efficiency of 5G transport networks. Its ability to handle high-capacity traffic, accommodate diverse traffic types, and efficiently aggregate data streams makes it an integral component in facilitating the seamless and robust operation of 5G networks, particularly within the backhaul and core network segments.

4.5.3 Optical Amplifiers

Erbium-Doped Fiber Amplifiers (EDFAs) play a crucial role in optical communication systems by enhancing the strength of optical signals transmitted through long-distance optical fibers. Here's how they function:

- **Signal Strength Enhancement:** EDFAs are designed to amplify optical signals that might weaken as they travel across extensive distances in optical fiber networks. Signal attenuation, caused by natural factors or transmission losses, can degrade the signal strength. EDFAs counteract this by boosting the power of these signals.

- **Overcoming Signal Attenuation:** As optical signals propagate through fiber optic cables over long distances, they experience attenuation, where the signal strength diminishes. EDFAs counter this attenuation by selectively amplifying the optical signals, ensuring they maintain adequate strength to preserve signal integrity and quality.
- **Maintaining Signal Quality:** By amplifying the optical signals, EDFAs help maintain the quality of transmitted data over extended distances. This is crucial in ensuring that the information carried by these optical signals remains intact and undistorted, facilitating reliable and high-quality communication.

Overall, EDFAs serve as vital components within optical communication systems, playing a pivotal role in overcoming signal attenuation and maintaining the strength and integrity of optical signals traversing extensive distances in fiber optic networks.

4.5.4 Optical Cross-Connects (OXCs)

Optical cross-connects play a key role in managing and routing optical signals within a network. Here's how they contribute to network flexibility and agility:

- **Flexible Signal Routing:** Optical cross-connects enable the dynamic and flexible routing of optical signals within the network infrastructure. They offer the capability to establish, modify, or terminate connections between various optical paths without the need for physical changes to the fiber connections.
- **Efficient Network Management:** By allowing operators to establish new connections or reroute traffic as needed, optical cross-connects streamline network management. Operators can adapt to changing traffic patterns or requirements without undergoing extensive physical rewiring, significantly reducing operational complexities.
- **Enhanced Agility:** The flexibility provided by optical cross-connects enhances network agility. Network operators can respond promptly to changing demands, traffic fluctuations, or network maintenance requirements by rerouting optical signals in a dynamic and efficient manner.
- **Optimizing Resource Utilization:** These cross-connects facilitate the optimal utilization of network resources by enabling efficient allocation and redirection of optical signals. This capability ensures that network capacity is utilized effectively, enhancing overall network performance.

In essence, optical cross-connects serve as crucial components in modern optical networks, providing the agility and flexibility necessary to efficiently manage and route optical signals without the need for physical modifications to the fiber connections. This capability is fundamental in meeting evolving network demands while ensuring efficient resource utilization and streamlined network operations.

4.5.5 Packet Brokers

Packet brokers serve as essential components within network analysis and monitoring systems. Here's how they contribute to the efficiency of network traffic analysis:

- **Traffic Aggregation:** Packet brokers gather and aggregate network traffic from various sources within the network. They consolidate diverse streams of data, ensuring that all relevant packets are available for analysis and monitoring.
- **Filtering and Packet Optimization:** These tools filter incoming network traffic based on predefined criteria, including protocols, addresses, or specific data patterns. By selectively forwarding the required packets to designated monitoring tools, packet brokers optimize the data flow. This filtering process ensures that monitoring solutions receive only the pertinent packets, reducing the load on monitoring tools and improving overall efficiency.

- **Load Balancing:** Packet brokers distribute the filtered packets evenly among different monitoring tools or systems. This load balancing ensures that no single monitoring tool is overwhelmed with excessive data, maintaining optimal performance across the monitoring infrastructure.
- **Enhancing Monitoring Tools' Effectiveness:** By providing precisely the right packets to each monitoring tool, packet brokers enable these tools to focus on relevant data for analysis and troubleshooting. This targeted approach enhances the effectiveness of monitoring solutions, allowing for more accurate and efficient network analysis.
- **Troubleshooting and Security:** Efficient distribution of relevant packets to monitoring tools allows for rapid identification and resolution of network issues or security threats. Packet brokers facilitate the timely delivery of crucial data to monitoring and analysis systems, aiding in swift troubleshooting and threat mitigation.

In essence, packet brokers streamline the flow of network traffic to monitoring tools, ensuring that these tools receive optimized and relevant packets for analysis. This selective distribution enhances the efficiency of network analysis, troubleshooting, and security monitoring within complex network environments.

4.6 Introduction to Mobile Transport Networks

Mobile transport networks are an important component of modern telecommunications infrastructure. They facilitate the movement of voice, data, and multimedia traffic between mobile devices and the core network. These networks are responsible for the seamless transmission of information across various technologies, such as 2G, 3G, 4G, and 5G, enabling wireless communication on a global scale.

Key components of mobile transport networks include base stations (cell towers), backhaul connections, and essential network elements. Base stations act as access points for mobile devices, providing radio coverage in specific geographic areas. Backhaul connections link these base stations to the core network, typically using fiber optics or microwave links.

Mobile transport networks must meet the growing demand for high-speed data transfer, low latency, and reliability. As new generations of wireless technology, like 5G, emerge, these networks continue to evolve to support faster data rates, increased capacity, and support for emerging technologies like IoT and edge computing. This evolution involves the deployment of advanced technologies such as small cells, beamforming, and network slicing to enhance performance and efficiency.

The backhaul and backbone structure in a Mobile Network Operator's (MNO) 5G network is a crucial part of the infrastructure responsible for connecting various network elements, including base stations, core network components, and data centers. Here's an overview of the 5G backhaul and backbone MNO structure:

- **Fiber-Optic Backbone:** The backbone of a 5G network is typically built on a high-capacity fiber-optic network. Fiber-optic cables provide the necessary bandwidth to handle the massive data traffic generated by 5G networks and support low-latency communication.
- **Core Network:** The core network is the central part of the MNO's infrastructure, and it serves as the heart of the 5G network. It includes elements like routers, switches, and data centers. The core network manages tasks such as call routing, data routing, and network functions.
- **Edge Data Centers:** In 5G networks, edge data centers play an important role in reducing latency for applications that require real-time processing. These data centers are strategically located closer to the users to enable faster response times and improved user experiences.
- **Microwave and Millimeter-Wave Backhaul:** While fiber-optic backhaul is the preferred choice for high-capacity 5G networks, microwave links and millimeter-wave technology may be used in areas where laying fiber is challenging. These wireless backhaul solutions provide an alternative means of connecting base stations.

- **Small Cell Backhaul:** In dense urban areas, small cells are deployed to enhance 5G coverage and capacity. Fiber-optic connections are used as backhaul for these small cells to ensure high-speed and low-latency communication.
- **Network Slicing:** Network slicing allows for the creation of multiple virtual networks within the same physical infrastructure. The backbone and core network must support network slicing to efficiently manage and allocate resources for different 5G use cases.
- **IP/MPLS Networks:** Internet Protocol/Multi-Protocol Label Switching (IP/MPLS) networks are commonly used in the backbone to provide scalable and efficient routing and traffic management in 5G networks. [32]
- **Redundancy and Resilience:** Backup links, fail-over mechanisms, and redundancy are essential in the backbone and backhaul networks for minimizing service disruptions and to ensure network availability and reliability.
- **Security Infrastructure:** Robust security measures, including firewalls, intrusion detection systems, and encryption, are crucial in the backbone and core network to protect against cyber threats and ensure the confidentiality and integrity of data traffic.
- **Network Management and Orchestration:** Advanced network management and orchestration systems are used to monitor and control the entire network. These systems help optimize network performance, troubleshoot issues, and dynamically allocate resources.
- **Regulatory Compliance:** MNOs must adhere to regulatory requirements related to 5G spectrum licensing, data privacy, and security standards in both the backbone and backhaul networks.

The 5G backhaul and backbone structure form the backbone of the MNO's 5G network, providing the necessary infrastructure to deliver high-speed, low-latency, and reliable communication services to users. This infrastructure is continually evolving to meet the demands of a highly connected and data-intensive world.

4.6.1 Sequence of Deployment of a Mobile Transport Network

In a mobile network, the backbone network typically comes first before the backhaul network.

The sequence of development for a mobile network usually is as follows:

1. **Backbone Network:** Typically, the backbone network is established as the initial step as the core infrastructure of the mobile network. It connects key elements of the network, such as Mobile Switching Centers (MSCs), data centers, and other core network components. The backbone network provides high-capacity and long-distance connectivity between different regions or cities, forming the foundation for the entire mobile network. This core infrastructure connects major points of presence (PoPs) such as data centers, internet exchange points (IXPs), and central offices. It uses high-capacity links, often fiber optic cables, for long-distance data transmission.
2. **Access Network:** After setting up the backbone network, the focus shifts to building the access network, which includes cellular towers (base stations) and other access points. These access points are distributed throughout the coverage area to provide mobile service to users. The access network is the part of the network where mobile devices connect and communicate with the network.
3. **Backhaul Network:** Once the access network is in place, the backhaul network is deployed to connect the access points (cellular towers) to the core network (backbone). The backhaul network is essential for aggregating the traffic from numerous local access points and routing it efficiently to the core network. It can use various technologies like fiber optic cables, microwave links, or other high-capacity connections. The backhaul network connects local access points, such as cellular towers, DSLAMs (Digital Subscriber Line Access Multiplexers), and Wi-Fi hotspots, to the core network (backbone). It ensures efficient data aggregation and routing of traffic from local access points to the backbone network. Backhaul networks can use various technologies, including fiber optics, microwave links, or even satellite connections, depending on the geography and requirements of the specific area being served.

The sequence described here ensures that the backbone network is established first to support long-distance data transmission. Subsequently, the backhaul network is built to extend this connectivity to local access points, ensuring that users and devices can efficiently connect to the core network and access telecommunication services. However, in practice, network development strategies may vary based on factors such as geographic considerations, population density, and existing infrastructure.

Acquiring sites and rights of way for fiber installations is costly and time-consuming, and one possibility is that transport - especially physical resources like fiber but possibly also high-layer operations - is shared in such a way that one entity owns and operates the network. This entity can be a neutral host, meaning a third-party 'independent entity' offers services to multiple tenants.

4.7 Functionalities

3GPP (3rd Generation Partnership Project) plays a crucial role in defining standards and functionalities for 5G backhaul and backbone networks.

3GPP's functionality for 5G backhaul and backbone networks is about establishing standards and specifications to make sure these networks are capable of supporting the diverse and demanding requirements of 5G services while maintaining security, efficiency, and flexibility.

It sets the global standards for mobile communications, including 5G, and defines the specifications and requirements for the backhaul and backbone networks that support 5G services, ensuring interoperability and compatibility among different vendors' equipment.

4.7.1 3GPP Functionalities

Here's an overview of the 3GPP functionality for 5G backhaul and backbone networks:

- **Enhanced Mobile Broadband (eMBB) Support:** 3GPP standards for 5G backhaul and backbone networks include support for eMBB services, which are designed to deliver high-speed, high-capacity data connections for applications like ultra-HD video streaming and virtual reality.
- **Low Latency and Reliability:** 5G backhaul and backbone networks are expected to provide low-latency connections, essential for applications like autonomous vehicles and industrial automation. 3GPP standards define requirements for low latency and high reliability in these networks.
- **Network Slicing:** 3GPP introduces the concept of network slicing, allowing operators to create multiple virtual networks within a single physical infrastructure. This functionality enables customized network services for different use cases and applications, all sharing the same backhaul and backbone resources.
- **Support for Massive IoT (mIoT):** 3GPP standards for 5G backhaul and backbone networks also accommodate massive IoT deployments, which involve connecting a vast number of low-power, low-data-rate devices. This functionality ensures efficient resource utilization and scalability for IoT applications.
- **Quality of Service (QoS) and Traffic Prioritization:** 3GPP defines QoS mechanisms that enable operators to prioritize traffic based on application requirements. This ensures that important applications receive the necessary network resources to maintain a high quality of service.
- **Network Function Virtualization (NFV) and Software-Defined Networking (SDN):** 3GPP incorporates NFV and SDN concepts, allowing for dynamic and flexible network management. NFV enables the virtualization of network functions, while SDN allows for programmable and centralized control of network resources.
- **Security Enhancements:** 3GPP specifies security features to protect the backhaul and backbone networks from cyber threats and unauthorized access. These features include authentication, encryption, and network monitoring.

- **Multi-Connectivity:** 3GPP standards support multi-connectivity, where a user device can simultaneously connect to multiple cells or access points. This improves reliability and performance by utilizing multiple network resources.
- **Coordination with Radio Access Networks (RAN):** The 3GPP standards ensure coordination between the backhaul and backbone networks and the 5G radio access networks (RAN) to optimize overall network performance.

3GPP plays a crucial role in defining the functionality and standards for 5G backhaul and backbone networks, ensuring that they meet the diverse requirements of 5G services and applications, including high-speed data, low latency, and support for IoT and important services. These standards help in creating a robust and efficient infrastructure to support the 5G ecosystem.

Relevant 3GPP Interfaces:

- S1-U: User plane interface between the LTE network and the core network
- NG-U: User-plane Interface between the gNB and the CN, similar to the S1 interface of the LTE
- NG-C: Control-plane Interface between the gNB and the CN, similar to the S1 interface of the LTE
- Xn: Interface between gNB and gNB within the same PLMN
- Uu: Wireless interface between the terminal and the 5G access network
- SCTP Interface: Transport layer protocol used for signalling

4.7.2 Operation Administration Maintenance Functionalities

The 5G transport network consists of both path and section layers, each having its own OAM (Operation Administration Maintenance).

The path OAM functions shall support the following:

- **Network Monitoring:** OAM tools continuously monitor the 5G network to detect issues, performance bottlenecks, and faults in real-time. This monitoring covers various network components, including base stations, essential network elements, and transport infrastructure.
- **Fault Detection and Diagnosis:** 5G OAM helps identify and diagnose network faults, errors, or failures promptly. It provides alerts and notifications to network administrators when issues arise.
- **Performance Management:** OAM processes track and analyze the performance of 5G networks, ensuring that they meet the defined quality of service (QoS) and key performance indicators (KPIs). This includes measuring data rates, latency, and reliability.
- **Configuration Management:** OAM tools facilitate the configuration of network elements, such as adding or removing base stations, updating parameters, and optimizing resource allocation for efficient network operation.
- **Security Management:** Ensuring the security of 5G networks is an important aspect of OAM. It involves implementing security policies, monitoring for security threats, and responding to security incidents.
- **Software Updates and Upgrades:** OAM processes manage the deployment of software updates, patches, and upgrades to network components to ensure they are up-to-date and secure.
- **Proactive Maintenance:** OAM aims to minimize downtime by allowing network operators to schedule maintenance activities during low-traffic periods and perform proactive maintenance to prevent issues.
- **Troubleshooting and Root Cause Analysis:** When network problems occur, OAM tools assist network engineers in troubleshooting and identifying the root causes of issues, which helps in faster resolution.
- **Resource Optimization:** OAM processes help optimize network resources, such as spectrum, bandwidth, and power, to improve network efficiency and reduce operational costs.
- **Vendor Interoperability:** As 5G networks often consist of equipment from different vendors, 5G OAM plays a role in ensuring interoperability between these vendor-specific elements.

- **Service Assurance:** OAM also includes service-level monitoring to ensure that the services delivered over the 5G network meet their performance and availability targets.

The section layer OAM functions can provide the performance supervision and management, and should include the following functions:

- Continuity supervision
- Connectivity supervision
- Error performance monitoring
- Remote defect and error reporting
- Delay measurement (in-service and out-of-service)
- Protection communications channel
- Client signal fail indication

Overall, 5G OAM is essential for maintaining the reliability, performance, and security of 5G networks, especially considering the complexity and scale of these networks. It involves a combination of automated tools, network management systems, and human expertise to keep 5G services running smoothly.

5 Risk Scenarios

This chapter provides the current view on a range of risk scenarios. It deals with an estimation of probabilities that each of this risk scenarios could become reality. These estimations are based on detailed discussions of potential scenarios of weaknesses or attacks. These discussions conclude with an assessment of the probability of occurrence.

The following table shows if the transport network of a 5G mobile network is assessed as a relevant subsystem of each risk scenario.

Perspective	Section	Risk Scenario	Transport	CIA
Public MNO Perspective	5.2.1	Emergency Call (TKG § 164)	x	A
	5.2.2	Cell Broadcast (TKG § 164a)	x	A
	5.2.3	Telecommunications Secrecy (TTDSG § 3)	x	C
	5.2.4	Emergency Preparedness (TKG § 184)	x	A
	5.2.5	Regional Network Failure	x	A
	5.2.6	Multi-Regional Network Failure	x	A
	5.2.7	Failure of Mobile Service	x	A
	5.2.8	Fraud		I
	5.2.9	Subscriber-Data Leak	x	C
User Perspective	5.3.1	Urgent Telephone Calls Not Possible	x	A
	5.3.2	Authenticity of the Receiver of a Call		I
	5.3.3	Integrity of Transmitted Data	x	I
	5.3.4	User's Identity		C
	5.3.5	Eavesdropping of a Communication	x	C
	5.3.6	Slowing Down/Restricting the Connection	x	A
	5.3.7	Connection Failure	x	A
	5.3.8	Tapping or Altering of the Geographical Position		C
Private 5G Network Operator Perspective	5.4.1	Network Failure	x	A
	5.4.2	Degraded Services/Connections	x	A
	5.4.3	Information Disclosure	x	C
	5.4.4	User Identities (Authenticity/ Integrity)		I

5.1 Attack Paths and Risk Sources

For each risk scenario a couple of different attack paths and risk sources will be discussed within this document. The result of the discussion will be a probability level for the impact on the risk scenario.

The term attack path refers to a sequence of steps or actions that an attacker can take to exploit vulnerabilities in a system or network, leading to security issues. It outlines the route or method an attacker follows to achieve his goals, often involving a series of interconnected actions that exploit weaknesses in a system's defenses.

A risk source is defined as the origin or cause of a potential risk or threat. A risk source is identified as the factor or element that can lead to vulnerabilities or hazards.

As many of the attack paths (or at least sub-sequences of the attack paths) and risk sources are similar for most of the risk scenarios, each attack path and risk source will be introduced from a **transport network** perspective in this chapter.

5.1.1 Misconfiguration (AP1)

Misconfiguration stands as a noteworthy risk source in 5G transport networks. In the context of transport networks, errors in configuring devices, protocols, or network elements can lead to performance degradation or outage of parts of the network.

5.1.1.1 Probability Determinants

The probability of these occurring can be assessed based on several factors:

- **Human Error Frequency:** The likelihood of an inexperienced network administrator making a misconfiguration error could vary based on their training, experience level, and familiarity with the system.
- **Complexity of the Task:** The complexity and potential for error increase with the complexity of the task.
- **Quality of Processes and Controls:** If there are inadequate checks, balances, or verification steps in the change management process, the probability of errors might increase.
- **Frequency of Changes:** If there are frequent changes, it might increase the probability of error due to the increased exposure to potential mistakes.
- **Training and Support:** The level of training and support provided to the network administrator could impact the likelihood of such errors.
- **Redundancy and Recovery Measures:** The presence (or absence) of backup configurations, rollback procedures, or fail-safes could affect the consequences and the likelihood of this particular error causing a significant impact.
- **System Complexity and Documentation:** The complexity of the 5G network infrastructure and the clarity of documentation might influence the likelihood of misconfiguration.
- **Equipment Failures:** Malfunctioning or faulty network hardware, including switches, routers, or other network components, can also lead to disruptions and loops within the network infrastructure.
- **Lack of Testing or Validation:** Inadequate testing or validation procedures before implementing routing policy updates may lead to misconfigurations going unnoticed, potentially causing disruptions.

- **Sudden Surge in Data Traffic:** Unforeseen spikes in data volume can overwhelm network capacity, causing congestion and inefficiencies in handling the increased load.
- **Internal Attack:** An internal attacker can exploit or deliberately create misconfigurations in routers to gain unauthorized access and launch an attack on the network.

5.1.1.2 Confidentiality (AP1-C)

In general, there is a vast number of parameters and options to configure when administrating a network. In particular, configurations that affect the confidentiality of transmitted data are of critical importance. These include unnecessary open ports, running services or privileges, improperly configured permissions, the use of default accounts and passwords, error log handling, security settings, misconfigured firewalls or firewall capabilities integrated in router equipment, and many more.

Especially regarding data confidentiality, improperly configured security settings might allow an attacker to access sensitive information. Examples for such misconfigurations are the use of deprecated cryptographic ciphers, for example weak cipher suites in Transport Layer Security (TLS) implementations such as TLS_RSA_WITH_AES_128_CBC_SHA [87], and improper key management, such as improper key reuse or irregular key updates [40]. Messages that are encrypted with weak or deprecated ciphers can be intercepted by an attacker and decrypted [70]. In order to mitigate the risk of misconfigured security parameters, network administrators can follow guidelines published by NSA [40] and NIST [41], that describe secure configurations for cryptographic schemes and key management.

As mentioned further above, misconfigurations also include exposure to the public internet, i.e. open ports, unwanted firewall behavior or unnecessary services. Transport network equipment, such as routers or switches, are important components of the 5G architecture and take care of managing incoming and outgoing data packets, which makes them an immediate connecting part between users and the network.

Unprotected exposure of hardware components of transport network equipment, such as routers or switches can pose severe vulnerabilities and open attack paths for network intrusion. By hijacking an unprotected router of the network, an attacker can steal sensitive data of customers [44]. Therefore, it is of utmost importance to review public exposure of hardware components to the internet and limit ports, privileges, and services to the required minimum.

5.1.1.3 Integrity (AP1-I)

Misconfigurations are common entry points for attackers to hijack a network component. An IBM Cyber Security Intelligence Index Report noted a percentage of 95% of all breaches to be connected to human errors [43], which also include administrative errors in network configurations. Once attackers have gained access to network components, they can alter data that passes through and tamper with established connections, violating the integrity. Misconfigured routers, switches or ethernet bridges, in particular, are prime targets of the transport network as they are pivotal in directing traffic. Attackers can exploit misconfigurations to reroute traffic through malicious servers (e.g. by outdated routing information), enabling man-in-the-middle attacks where data can be intercepted and modified. In case of security misconfigurations, attackers can also disable security features, such as firewalls and encryption protocols, making it easier to carry out further attacks and maintain persistent access. These actions compromise the integrity of the network, leading to significant data breaches and loss of trust in the affected systems.

5.1.1.4 Availability (AP1-A)

A major part of availability issues that result from misconfigurations stems from Denial of Service (DoS) attacks or data congestions. Inadequate firewall configurations of the GPRS Tunnel Protocol (GTP), which allows data transmission between different network interfaces, may cause data congestions and lead to service unavailability. Security for GTP was considered only after it was initially designed, making it prone to attacks and exploits [24]. Incorrect configurations for message filtering, exploit detection or message-

length control are common entry points for attacks on the protocol and enable DoS attacks, which in turn result in downtimes of the network service [45].

Furthermore, network unavailability can also be caused by data congestion or network loops. Due to incorrect routing configurations, packets may be stuck in a loop between two or more routers, amplify itself and lead to slow connections or network failures [28]. The main cause for these kinds of unavailability risks is router table configuration and physical link loops, but also active IP spoofing attacks [46]. In addition, availability issues include slowed down connections. For example, a misconfigured firewall that is configured to inspect any incoming and outgoing data adds latency to the network and impedes data throughput. Moreover, router misconfigurations regarding preferred DNS servers, VPN tunnels or resource limitations (usage of CPU, memory, and storage) can create a bottleneck in data throughput and slow down the connection.

5.1.2 Insufficient Access Control (AP2)

Insufficient access controls within 5G transport networks present a significant risk, allowing unauthorized entities or devices access to essential network components. The absence of robust access control mechanisms can have severe consequences. In this context, we differentiate between insufficient access control regarding virtual access through software interfaces or compromised network devices, and physical access via an intruder.

5.1.2.1 Probability Determinants

Several factors contribute to the risk scenario of Insufficient Access Controls in a 5G transport network:

Inexperienced Administration: Lack of experience or expertise in managing access controls increases the probability of misconfigurations or insufficient restrictions.

Complexity of 5G Networks: The intricate nature of 5G networks, involving various components and technologies, can make it challenging to implement and manage robust access controls effectively.

Rapid Technological Advancements: The rapid pace of technological advancements may result in evolving security threats and vulnerabilities, requiring constant updates and adaptations in access control measures.

Human Error and Oversight: Oversight or mistakes in configuring access controls, particularly in granting or revoking permissions, can lead to vulnerabilities and unauthorized access.

Inadequate Monitoring and Auditing: Lack of robust monitoring or auditing mechanisms can make it difficult to identify and rectify access control issues promptly.

Third-Party Risks: Vulnerabilities arising from interconnected systems or interfaces with third-party entities may introduce access control weaknesses.

Evaluating and addressing these factors is crucial in establishing robust access controls to mitigate the risks associated with insufficient access controls in a 5G transport network.

5.1.2.2 Confidentiality (AP2-C)

Bypassing access control mechanisms allows attackers to gain unintended privilege rights, break into accounts with elevated rights or even hijack the whole network. Privilege escalation is a known attack vector in identity management systems and highly relevant not only for virtualized services, such as cloud providers like AWS and Azure [48], but also for telecommunication components. Regularly, passwords for remote access to routers are not changed, leaving them with default passwords [88, 89]. This may result in unauthorized access and undesired activities in the network. An attacker that has privileged access to network components, can eavesdrop transmitted data.

Though unauthorized access often stems from misconfigured cloud servers, faulty software implementations or firewalls, the underlying problem is still insufficient access control. For example, in an attack on Capital One [59] a misconfigured firewall helped attackers generate fraudulent access tokens for the network service. These forged access tokens enabled the attackers to steal sensitive customer data.

Unprotected exposure of hardware components can pose severe vulnerabilities and open attack paths for network intrusion. By hijacking an unprotected router of the network, an attacker can steal sensitive data of customers [44]. Therefore, it is of utmost importance to review public exposure of hardware components to the internet and limit ports, privileges, and services to the required minimum.

As mentioned further above, insufficient access control includes the possibility to gain physical access to hardware components of the transport network, i.e. routers, switches, bridges, and controllers, but also any fiber optic equipment or cables. An attacker that has physical access to the transport cables can tap the fiber and eavesdrop the communication, thus breaching the confidentiality of the transmitted data [47]. Intrusion on optical fiber creates variations in signal power levels which can be detected, and signal reflections which can be localized by state-of-art network equipment [78].

5.1.2.3 Integrity (AP2-I)

The lack of reliable intrusion detection systems (e.g. by blocklisting forbidden users instead of allowlisting permitted users) may lead to man-in-the-middle attacks, where attackers intercept and alter messages. For example, the 5G-AKA protocol for Authentication and Key Agreement contains weaknesses in network access authorization, which allows attackers to spoof a device and send data (SMS messages and phone calls) in the name of another user, effectively stealing another identity. This vulnerability is caused by erroneous implementations of authentication mechanisms by MNOs, which do not obey to best-practice access control standards and transfer secret keys in an insecure manner while a device is roaming [51, 52].

Furthermore, attackers can change data or configurations within a 5G network slice and thus compromise the integrity of the network and the data processed in it. For example, insufficient access control in one network slice allows attackers to compromise this respective network slice as an entry point and then propagate across different interconnected network slices. Eventually, they can hijack an entire network infrastructure with multiple network slices, solely by exploiting a broken access control on a single slice. Consequently, attackers can manipulate traffic routing or change messages between users, which would lead to incorrect information or faulty data transfer [50].

5.1.2.4 Availability (AP2-A)

Physical access to transport network components poses a major risk for network availability. In fact, fiber optic cables with insufficient access control are not only exposed to wiretapping and eavesdropping, threatening the confidentiality of data, but also to damage caused by attackers or the influence of nature (e.g. natural disaster, heavy rain, etc.) [11]. For example, targeted attacks on fiber optic cables can cause a major unavailability of the entire network [49]. In general, the transport network is difficult to protect due to physical size of the network and related costs, and point-to-point links are more vulnerable against physical attacks as compared to switched links or ring topologies.

Insufficient software-side access control that lets attackers hijack the network may also cause a service unavailability. Vulnerabilities in 5G networks can cause DoS attacks and service outages, impacting the network's ability to deliver consistent and reliable connectivity and services to users. Once the attackers have gained access to a network component, they can move between routers and access points or access different parts of the network [50]. A recent cyber-attack took down over 600.000 routers in the US by executing pre-built DDoS attacks by means of injected hardware. It is assumed that initial access to the routers has been gained by weak credentials, e.g. default passwords or exposed administrative interfaces [67]. In general, it is of utter importance to reset default passwords for privileged (remote) access, as well as to secure any interfaces that can provide an entry point for attackers to gain administrative rights to the system. Attackers that gain access to a router can inject malware, which disrupts the functionality of the device.

5.1.3 Implementation Flaw (AP3)

The number of vulnerabilities which are discovered in the software of network equipment has increased drastically over the last years, for example shown by MITRE and NIST [37]. Such vulnerabilities and implementation flaws pose significant operational and security challenges within 5G transport systems, and can lead to various possible incidents in the following categories:

- **Network Outages**, impacting 5G service availability and reliability, leading to financial losses and tarnishing the operators' reputations.
- **Data Integrity**, potentially causing data corruption or unauthorized modifications during transmission.
- **Performance Degradation**: Implementation flaws can result in suboptimal network performance, causing increased latency, packet loss, and reduced throughput, directly impacting user experience.
- **Regulatory Non-Compliance**: Products with implementation flaws may fail to meet industry standards, which may cause legal penalties due to non-compliance with regulations.
- **Costly Remediation**: Identifying and rectifying flaws is resource-intensive, requiring software updates, patches, and potentially hardware replacements, incurring substantial time and costs.
- **Expanded Attack Surface**: Implementation flaws expand the attack surface, rendering network products more vulnerable to exploitation by cybercriminals and threat actors.

5.1.3.1 Probability Determinants

Several factors contribute to Implementation Flaws in a 5G transport network concerning the quality of the product:

Software Bugs or Defects: Errors or flaws within the software codes or algorithms used in 5G network components can lead to implementation flaws, affecting the functionality and reliability of the network.

Incomplete or Incorrect Configuration: Inaccurate configuration settings or incomplete setup during the implementation phase can result in operational inefficiencies or vulnerabilities.

Integration Issues: Challenges in integrating new 5G technologies or components with existing infrastructure may lead to compatibility issues, causing implementation flaws.

Faulty Hardware Components: Defective or substandard hardware components used in the network infrastructure can result in implementation flaws, affecting the network's performance and stability.

Testing Shortcomings: Insufficient testing procedures or inadequate testing environments during the implementation phase can fail to detect potential flaws or vulnerabilities in the network.

Lack of Compliance to Standards: Deviation from established industry standards or protocols during implementation can introduce flaws, impacting interoperability and security.

Legacy Equipment Limitations: Older equipment lacking nuanced access control capabilities presents an inherent risk by limiting the ability to restrict user privileges to specific commands.

Assessing and mitigating these factors is crucial to ensure a robust implementation process and minimize the risk of implementation flaws within a 5G transport network.

5.1.3.2 Confidentiality (AP3-C)

Data confidentiality is at high risk, when falsely implemented algorithms or products are used [19]. The most common case for data breaches are vulnerabilities in software implementations that have not been patched yet [54]. Security updates should be rolled out and installed regularly, especially when a security vulnerability is made public and is not fixed yet. In the worst case, an attacker can conduct a zero-day exploit, i.e. an exploit of a previously undiscovered vulnerability, or a one-day exploit, when a known vulnerability has not been patched yet.

Another common cause for a data breach is the use of old or outdated equipment with poor quality or expired security support service. The equipment might be too old for current state-of-art algorithms and use legacy algorithms instead [81]. In addition, the system might be prone to vulnerabilities that have been patched for newer systems only. For example, a study has shown that 98% of all victims of the WannaCry ransomware in 2017 were using the outdated operation system Windows 7 [55]. Such malware, injected through unpatched implementation flaws, may also aim at stealing data from infected components.

Furthermore, errors in software code, integration issues, and non-compliance with industry standards can lead to unauthorized data access. Lack of encryption and access controls in legacy equipment can compromise data confidentiality [56].

Common practice for transport networks is to use data plane encryption by means of symmetric cryptography such as AES-256 (low computational complexity, quantum computing secure), where the respective key exchange is implemented via OAM interfaces, well separated from user plane traffic [39].

5.1.3.3 Integrity (AP3-I)

Software bugs with exploitable vulnerabilities that allow man-in-the-middle attacks are a threat for data integrity. An attacker sitting between two communication endpoints can intercept and alter the messages. Most attacks on data confidentiality that stem from implementation flaws also affect data integrity, as attackers can not only read exposed confidential data but also modify and tamper with it.

Exploitation of unknown vulnerabilities, i.e. zero-day exploits, integration challenges, and testing shortcomings can lead to data corruption or unauthorized modifications. Deviation from established industry standards can further exacerbate these risks, leading to compromised data integrity. Additionally, poorly implemented encryption algorithms can be exploited to decrypt and re-encrypt data with malicious alterations. Attackers can exploit buffer overflow vulnerabilities in vulnerable router software to inject malicious code, altering the behavior of data transmission and corrupting or deleting data [86]. These implementation flaws create numerous entry points for attacks that compromise the integrity of data, undermining the trust and reliability of the affected systems.

5.1.3.4 Availability (AP3-A)

Flaws in the implementation of software or errors in the instantiation of network components can cause severe availability issues [6], [7]. Faulty and insufficiently tested software updates may cause network outage [66], in particular when involving multi-vendor deployments. Faulty SW updates were the single largest cause of telecom user hours lost in the EU in the year 2022 [38].

Also flawed network dimensioning or flawed network architectures may lead to degraded network performance or network outage. Examples are insufficient link capacity, insufficient number of redundant transport routes, or architectures enabling alien wavelength attacks in optical networks [78].

5.1.4 Specification Weakness (AP4)

The spectrum of specification weaknesses across various network protocols in a transport network presents a multifaceted scenario. Assessing their vulnerability probabilities:

- Ethernet might pose a low to medium probability of vulnerabilities due to common risks like spoofing.
- IPv6 weaknesses could range from medium to high due to complexities in migration and configuration issues [33].
- MPLS vulnerabilities may hover between low to medium probabilities, considering stringent configurations required for exploitation.
- GTP/UDP/TCP/IP protocols might exhibit a medium probability due to specific susceptibility.
- S1AP/SCTP/IP, X2AP/SCTP/IP may pose a lower to medium probability due to established mobile network security practices.

- SyncE, PTP, BGP, TLS/SSL, PIM, RSVP-TE, GMPLS, LDP, PWE3 could vary between low to medium probabilities based on specific vulnerabilities and network configurations [76].

These estimates are indicative and subject to variations based on network settings, security measures, and potential risks. Regular assessments and proactive security practices are pivotal in mitigating these weaknesses and enhancing network resilience [34].

5.1.4.1 Confidentiality (AP4-C)

Ethernet networks may suffer from confidentiality breaches when attackers use media access control (MAC) address spoofing to intercept data packets. This allows unauthorized access to sensitive information being transmitted over the network, potentially leading to data theft or eavesdropping [29].

Although MPLS is designed for efficient data routing, improper security measures can leave it vulnerable to traffic analysis and interception [33]. Attackers could potentially monitor and capture sensitive data as it traverses the MPLS network, compromising confidentiality.

Protocols like GTP, UDP, TCP, and IP can be exploited through techniques such as IP spoofing and session hijacking. These attacks enable malicious actors to intercept and access confidential information being transmitted across the network, breaching data privacy.

In BGP, attackers can perform route hijacking to redirect network traffic through malicious routes [71]. This unauthorized redirection allows them to intercept and potentially access sensitive data, thereby compromising the confidentiality of the communications.

In TLS/SSL protocols, man-in-the-middle attacks pose significant risks to confidentiality. An attacker intercepts the communication between two parties, decrypting and accessing sensitive data without authorization. They can also force a connection to use a less secure version of the protocol, weakening encryption and making it easier to compromise the confidentiality of the transmitted data [93, 94].

5.1.4.2 Integrity (AP4-I)

In general, the specifications for mobile telecommunication, and so for 5G specifically too, are a product of standardization research and development for several years and therefore are a robust foundation for the security of 5G. However, it is of utmost importance to check and update current state-of-art specifications for 5G transport network architecture and equipment. Regarding data integrity, the use of insecure message authentication or digital signature techniques can violate data integrity. In particular, newly added protocols and features can bring vulnerabilities into the specification, as real-life testing often uncovers previously unknown vulnerabilities. For example, the introduction of network virtualization emphasized security concerns regarding the virtualization of network functions and the shared use of routers and further communication hardware [90].

5.1.4.3 Availability (AP4-A)

The most prominent attack on availability of a network coming from specification weakness is a DoS attack or other resource exhaustion techniques. When the specification does not explicitly provide guidelines to prevent DoS attacks, certain implementations allow attackers to flood the network with e.g. MPLS labels and disrupt the service. In addition, there are several control plane attacks targeting protocols like LDP (Label Distribution Protocol) and RSVP-TE, which also impact the network availability. RFC 5920 “Security Framework for MPLS and GMPLS Networks” describes multiple security threats coming from insufficiently protected protocol specifications [65].

With respect to microwave transport, a looming availability issue is that the radio frequencies used by access systems tend to increase and compete with frequencies traditionally used for microwave backhaul. In the longer term, this may lead to shortages of available frequency resources for microwave backhaul, e.g., the 6 GHz band widely used for long haul overlaps with the 5925 to 6425 MHz unlicensed band allocated in 2020 (ECC Decision (20)01) and it is a contender for 6G access [26], [27].

5.1.5 Failure of Power Supply (AP5)

The reliability of power supply is an important factor in sustaining 5G transport networks. A power supply failure at a data center or network hub housing a key router can have diverse causes, such as a grid power outage, malfunctioning backup power systems, or defective power distribution equipment.

Power failures can trigger complete or partial shutdowns of crucial network infrastructure, rendering 5G services inaccessible to customers. Inadequate backup systems in essential network components during power outages can lead to data loss, including operational data, customer information, and configuration settings. Restoring power and network services post a power failure is time-consuming, particularly in cases of extensive damage, prolonging service disruptions.

5.1.5.1 Probability Determinants

Several factors contribute to the occurring of Failure of Power Supply in a 5G transport network:

Redundancy Measures: Assessment of redundancy in power systems to determine if backup power sources are adequately in place to prevent complete power loss in case of a primary power supply failure.

Maintenance Protocols: Evaluation of maintenance schedules and practices to ensure regular inspections, testing, and upkeep of power supply systems to prevent unforeseen failures.

Capacity Planning: Analysis of power supply capacities against network demands to ensure that the power infrastructure can handle peak loads without failures or overload-induced breakdowns.

Emergency Response Plans: Review of contingency plans and emergency response protocols to minimize downtime in the event of power supply failures and quickly restore network operations.

Monitoring and Alarming Systems: Assessment of monitoring tools and alarm systems to detect fluctuations, anomalies, or potential failures in the power supply, enabling proactive measures before a complete breakdown.

Fault Tolerance: Examination of the network's fault-tolerant design and measures in place to ensure that power supply failures in specific areas don't disrupt the entire network's functionality.

Environmental Considerations: Evaluation of environmental factors that could impact power supply systems, such as weather events, temperature extremes, or other external influences.

Assessing these factors helps in understanding the vulnerabilities and strengths of the power supply infrastructure within the 5G transport network, aiding in the mitigation and management of potential risks associated with power supply failures. [3]

5.1.5.2 Confidentiality (AP5-C)

Attacks on data confidentiality resulting from power supply failure could not be identified.

5.1.5.3 Integrity (AP5-I)

Attacks on data integrity resulting from power supply failure could not be identified.

5.1.5.4 Availability (AP5-A)

The network service is highly dependent on a stable and uninterrupted power supply, provided by a power grid. Abrupt and unforeseen power outages can lead to immediate disruptions of established connections and network service unavailability [12], [13]. Such failures not only affect the immediate availability of the network service but can also affect the operability of the whole infrastructure and lead to systemic failures that can take a significant amount of time and resources to resolve. Furthermore, power supply failures cause regional and multi-regional service outages, not only impeding important calls for private and

commercial users, but also influencing the communication of public authorities, first aid organizations or cell broadcasts in acute emergency scenarios [63], [64].

Additionally, insufficient emergency power supply can cause operational interruptions, impair the availability, and impede the re-instantiation of the service after a power outage. Abrupt cutting of power supply damages the hardware and can lead to data loss and corruption, operating system issues and file system damage. Power fluctuations and surges during such events can further exacerbate the problem, leading to hardware failures and reduced lifespan of critical components like optic fiber cables. The lack of redundant power systems means that a single point of failure can cascade into widespread service disruptions. Moreover, the recovery process often involves extensive diagnostics and repairs, which can extend downtime significantly, especially in case fiber cables are not easily reachable (e.g. underwater cables) [62].

A localized failure of power supply may disable an aggregation router deployed at or close to a radio cell site. Such a failure may cause one or a small number of radio cell sites to fail. Failure of the 5G transport network over a wide area may be caused by a failure of power supply on a (multi-) regional level or if affecting one or several data centers further away from the radio sites hosting routing/switching and/or wide area transmission equipment.

For example, in 2019, a lightning strike in the UK led to a power outage that affected 1 million people for about 40 minutes, disrupting telecommunications and emergency call services [77, p13]. In 2021, a massive power outage in Europe affected several countries, including France and Italy. The power outage was caused by a grid disconnection that led to significant disruptions to telecommunications services, including the unavailability of cell broadcast systems used for emergency alerts [85].

5.1.6 Failure of Hardware Components or Unavailability of Spare Parts (AP6)

Hardware component failures and the unavailability of spare parts can negatively impact the transport network of a mobile network. The following outlines the key facets of this risk scenario:

Failures in crucial hardware components such as routers, switches, and optical equipment can trigger service interruptions. This potential unavailability of 5G services to customers may cause inconvenience and potentially lead to financial setbacks for network operators.

Inadequate backup systems or redundancy in essential network components can result in data loss due to hardware failures. This could encompass the loss of operational data, customer information, and configuration settings.

The availability of spare parts is pivotal for timely repairs and maintenance. Disruptions in the supply chain or the obsolescence of equipment may lead to challenges in obtaining specific spare parts, thereby prolonging downtime and compromising network resilience.

5.1.6.1 Probability Determinants

Several factors contribute to assessing the risk scenario of Failure of Hardware in a 5G transport network:

Hardware Redundancy: Evaluation of redundancy measures in place to minimize the impact of hardware failures, including spare equipment, backup systems, or failover mechanisms.

Equipment Aging and Lifecycle Management: Assessing the age and lifecycle status of hardware components to predict potential failures and plan timely replacements or upgrades.

Fault Monitoring and Predictive Maintenance: Implementation of robust fault monitoring systems and predictive maintenance protocols to identify hardware issues before they escalate into failures.

Vendor Reliability and Support: Analysis of the reliability of hardware vendors, their track record in producing dependable equipment, and the availability of timely support and replacements.

System Resilience: Examination of the network's design for resilience against hardware failures, including the ability to reroute traffic, load balancing mechanisms, and redundancy in essential network components.

Operational Controls: Review of operational controls and procedures in managing hardware failures, such as backup configurations, disaster recovery plans, and quick replacement processes.

Environmental Considerations: Assessment of environmental factors impacting hardware, such as temperature controls, dust management, and protection against physical damage.

Testing and Validation: Regular testing and validation procedures to ensure hardware components function within specified parameters and can withstand potential stress factors.

Considering these factors aids in understanding vulnerabilities, implementing preventive measures, and developing contingency plans to mitigate the impact of hardware failures within the 5G transport network.

The exact consequences of the hardware failure depend on the role of the affected hardware in the network. Consider the following example:

In the network infrastructure a node processor board malfunctions. To address this issue, an expert was enlisted to replace the faulty board. However, during the replacement process, a significant misstep occurred: the replacement board was installed without a prior configuration backup, requiring the expert to configure it from scratch.

Adding to the challenge, the expert inadvertently disabled the initially functional board during the reconfiguration attempt. This unintended action led to the complete loss of visibility within the monitoring server, rendering it non-operational. As a consequence, the network operator was deprived of access to essential monitoring and control functions for an unspecified period.

The repercussions of this sequence of events can be substantial. With the monitoring server shut down, the operator faces an abrupt loss of real-time insights into network performance, status, and potential issues. This loss of visibility disrupts ongoing operations and hinders quick identification and resolution of emerging network issues.

The prolonged absence of access to the monitoring server poses significant challenges for the operator in responding promptly to potential network disturbances or outages. Urgent remedial actions are necessary to restore access and regain control over the monitoring system, ensuring the resumption of efficient network management and troubleshooting capabilities.

Considering the advancements in network redundancy, fault tolerance mechanisms, and improved expertise in handling hardware replacements, the probability in this case causing significant disruptions and prolonged loss of visibility within the monitoring server in the recent network context might lean more towards a medium level. However, this assessment depends on the specific network's resilience measures, backup protocols, and the efficiency of emergency response plans implemented by the network operator.

5.1.6.2 Confidentiality (AP6-C)

Attacks on data confidentiality resulting from failures of hardware components could not be identified.

5.1.6.3 Integrity (AP6-I)

Attacks on data integrity resulting from failures of hardware components could not be identified.

5.1.6.4 Availability (AP6-A)

A prominent hardware component failure risk is insufficient redundancy. Generally, backup systems ensure that the impact of a hardware component failure does not severely impact the availability. However,

unforeseen failures and physical attacks on networks with insufficient redundancy can lead to major unavailability issues if redundancy is not correctly implemented or one or more redundant entities are damaged. A real-world example is the intentional destruction of hardware components, such as fiber optic cables for the transport network [49].

Further availability issues can be caused by fiber optic interruptions, extreme wear or age of cables, routers and switches or external factors such as extreme weather conditions, resulting in longer outages. In general, all hardware components in transport networks, such as optical equipment, ethernet bridges, IP/MPLS routers and SDN controllers are susceptible to the mentioned failure risks. It is therefore of utmost importance to verify the functionality of hardware components and exchange them on a regular basis to prevent unforeseen failures. Additionally, the lack of proactive maintenance and monitoring can exacerbate these risks, as small issues may go unnoticed until they cause significant disruptions. Ensuring that there are comprehensive monitoring systems in place can help detect early signs of hardware degradation. Furthermore, implementing robust disaster recovery plans that include quick replacement and failover strategies is critical to maintaining high availability. Without these precautions, the 5G network can face prolonged downtimes, which can disrupt critical services, affect business operations, and lead to substantial financial and reputational losses [61].

A hardware failure disabling an aggregation router deployed at or close to a radio site may cause one or a small number of radio sites to fail. Failure of the 5G transport network over a wide area likely requires multiple simultaneous hardware failures at more central sites of the network, e.g. at regional data centers, as a good amount of redundant transport routes and equipment can be assumed in the backbone of the transport network.

Lastly, supply chain disruptions and the unavailability of spare parts can severely impact the availability of a network service. When a critical hardware device fails and cannot be promptly replaced due to logistical issues, bottlenecks in the supply chain or parts unavailability, it can lead to extended downtime for the affected network components. This downtime may result in service interruptions for users relying on that network, affecting productivity and customer satisfaction [91, 92].

5.1.7 Miscellaneous

5.1.7.1 Radio Interference

Attackers disrupt backhaul connections between cell towers and data centers by intentionally generating radio interference within microwave or millimeter-wave frequencies. This interference directly hampers the transmission of data in the mobile network.

To counter such threats, the design of wireless backhaul links must prioritize resilience and incorporate robust monitoring capabilities to promptly detect and respond to any interference. The probability in this case is somewhat dependent on the vulnerability of the wireless backhaul links and the sophistication of the attackers. Given the potential impact on the transferred data, this threat should be considered medium.

5.1.7.2 Environmental Weakness

Network equipment exhibits sensitivity to environmental factors, impacting its operational integrity:

- **Power Variations:** Altering power levels in equipment can lead to device malfunction or failure. Each vendor's equipment responds uniquely to power fluctuations, necessitating careful management to prevent damage or disruption.
- **Temperature Fluctuations:** Variations in temperature can affect signal quality, impacting data transmission reliability.
- **Reset Limits:** Network equipment imposes reset limits, restricting the number of allowable resets. Frequent resets beyond this threshold can compromise the equipment's stability and longevity, potentially leading to operational disruptions or failures.

Maintaining optimal conditions for network equipment involves mitigating power fluctuations, ensuring temperature control to minimize signal interference, and adhering to reset limitations to preserve equipment reliability and performance.

Given the ongoing advancements in technology and the efforts directed towards improving equipment resilience, the probability of environmental sensitivity in modern transport network equipment is moderately reduced. However, it is crucial to note that despite these advancements, factors like power variations, temperature fluctuations, and reset limits continue to pose potential risks to equipment integrity. Therefore, the probability of encountering environmental sensitivity in network equipment could be considered at a medium level in today's transport networks [21].

Concerning microwave transport for 5G networks, often high frequencies in the range of 71-86 GHz, known as the E-Band, are used, particularly in urban areas [57, 58]. Although E-Band wireless links provide a low-latency and high throughput medium, these frequencies are highly susceptible to weather anomalies like heavy rain and oxygen fluctuations in the transmission path. In addition, buildings and other infrastructures in the signal path reflect and scatter the signals, which is why most E-Band links are capped to only few kilometers. In general, physical signal paths are prone to environmental conditions and can cause unforeseen outages that disrupt the network service.

5.1.7.3 Energy Management

In a 5G transport network, dedicated components may manage energy across various telecommunication nodes. Recent concerns have surfaced about potential vulnerabilities within these components, primarily concerning their energy management capabilities. Experts fear that these vulnerabilities might be exploited, leading to disruptions in telecommunications operations.

The components' pivotal role in managing energy within the network makes them potential targets for exploitation. If a security breach occurs, it could result in various degrees of disruptions. For instance, a compromise could lead to localized service outages in specific network segments. In an extreme situation, such as a coordinated attack or a critical vulnerability exploitation, it might cause a partial or complete collapse of the transport network.

The probability of this scenario depends on several factors. The specific vulnerabilities within these components, existing security measures, the likelihood of exploitation by malicious actors, and the robustness of the network's overall security infrastructure are significant factors. Considering the essential nature of the component and the potential implications, the probability might be medium to high, given the recent concerns and heightened focus on security in telecommunications infrastructure [22].

5.2 Public Mobile Network Operator (MNO) Perspective

This section discusses the probability assessment of the risk scenarios related to the MNOs perspective.

For each subsection, a short evaluation of the relevance of the risk scenarios in the transport network is discussed and the probability that potential attack paths could be exploited is derived.

5.2.1 Emergency Call (TKG § 164)

While the radio access network (RAN) and core network predominantly handle emergency call services, the transport network (comprising backbone and backhaul elements) ensures seamless connectivity and robust communication within the overall 5G infrastructure. Though not directly involved in emergency call processing, disruptions or vulnerabilities within the transport network can indirectly affect the overall functionality and reliability of emergency services [4].

This assessment addresses the unavailability of emergency calls from a services perspective, i.e. an emergency call is assumed as unavailable if it cannot be performed via any of the networks present in the respective geographic area. From the perspective of a single MNO, similar assessment as for emergency preparedness (Section 5.2.4) applies.

Most geographical areas are covered by the networks of multiple mobile network operators. Sharing of some network assets is common among operators, in particular the sharing of cell sites or towers including the power supply and the backhaul cable ducts. More advanced network sharing implementations such as Multi-Operator RAN (MORAN) or Multi-Operator Core Network (MOCN) also share active network components in the RAN. In such case, also the sharing of transport network equipment, e.g. aggregation routers at the shared cell sites, can be assumed. In Germany, to the best of knowledge, active RAN sharing is confined to no more than two MNOs, and primarily deployed in sparsely populated areas. This implies that even if a particular shared radio site would fail, there is a high probability that at least one other radio site will cover the geographical area, either a radio site of the same MNO or a radio site of another MNO. Exceptions may be very sparsely populated areas or indoor locations having coverage from a very small number of cells only. Sharing of core network functions among the MNOs operating in a geographic area may pose a higher risk of unavailability of emergency calls, e.g. a cyber-attack on transport network equipment may disconnect the shared core network, but such core network sharing scenario is assumed to be confined to network operators with limited physical network assets (e.g. virtual network operators) and considered uncommon among the primary MNOs.

As mentioned, most geographical areas are covered by multiple MNOs and sharing of resources between multiple MNOs possesses the ability to reduce certain risks. Nevertheless to get an unbiased and implementation independent risk assessment, the following probability assessment will be done with the assumption of only one Public MNO existing. Unlike this simplified theoretical assumption of a single MNO, Germany benefits from having four MNOs, namely Telekom, Vodafone, Telefónica, and 1&1. Each of the MNOs independently operates its 5G Core, contributing to a robust and redundant 5G architecture. This redundancy becomes particularly crucial for the availability of emergency call services.

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Misconfiguration	Availability (AP1-A)	Given the above mentioned assumption, the risk of emergency calls being interrupted due to misconfigurations within a transport network is rated as MEDIUM . Under consideration of more than one MNO existing, this risk can be reduced to low, as in case of network misconfiguration the emergency call can most likely be delivered via the network of another MNO [79]. In addition, the probability for every available MNO network to simultaneously fail because of a misconfiguration is low.	Medium
Insufficient Access Controls	Availability (AP2-A)	The risk of unavailability of an emergency call due to insufficient access controls within a transport network is rated as low to medium. Disabling emergency calls requires to simultaneously interrupt the network of all MNOs in an area [79]. This may be achieved with moderate effort by disrupting the backhaul cables, for example, by attacking multiple radio sites in rural areas. Network sharing eases such attacks as backhaul cables of multiple MNOs may be shared or use the same cable duct. Simultaneous attacks on more centralized transport network equipment or cables of every available MNO network, e.g. cyber-attacks on transport equipment to disrupt the core networks of multiple MNOs simultaneously, would require plenty of resources and sophistication. Given the above assumption and explained probability range, the probability is rated as MEDIUM .	Medium
Implementation Flaw (Quality of the Product)	Availability (AP3-A)	The probability of unavailability of an emergency call due to implementation flaws is rated as MEDIUM . Due to the aspects described in the introduction of chapter 5.2, this risk can be reduced to low, as in case of an implementation flaw the emergency call can most likely be delivered via the network of another MNO [79]. For example, software errors in transport network components or defects in routers can lead to calls being misrouted instead of being connected to the nearest emergency call center [75]. However, the event that transport network equipment of every available MNO network is subject to software errors is rare.	Medium
Specification Weakness	Availability (AP4-A)	The probability of unavailability of the emergency call due to specification weaknesses in the 5G transport network is rated as LOW as the 5G transport network incorporates state-of-art transport protocols with robust QoS mechanisms defined by IETF.	Low
Failure of Power Supply	Availability (AP5-A)	Unavailability of an emergency call may be caused by a (multi-) regional power outage affecting the networks of all MNOs in the area. A regional power outage may be caused by human error, power system failure, theft of components, a natural disaster [8], [11], a terror attack, or sabotage [30], [75]. Given the variety of different sources for failures of power supply, their probability to be the cause of unavailability of an emergency call is rated as MEDIUM . For example, in 2019, a lightning strike in the UK led to a power outage that affected 1 million people for about 40 minutes, disrupting telecommunications and emergency call services [77, p13].	Medium
Failure of Hardware Components or Unavailability of Spare Parts	Availability (AP6-A)	The probability of a hardware failure to cause unavailability of an emergency call is rated as LOW , as the routed/switched transport connections most likely provide redundant hardware resources and transmission paths.	Low

5.2.2 Cell Broadcast (TKG § 164a)

Cell Broadcast is primarily an application layer service that utilizes the cellular network infrastructure, so the threat scenarios associated with it are more relevant to the broader cellular network rather than specifically the backhaul or backbone.

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Misconfiguration	Availability (AP1-A)	From the perspective of the transport network, cell broadcast data is equivalent to any other data transfer, i.e. any misconfiguration impairing data transfer would likely be detected during normal operations. However, cell broadcast traffic may be delivered with high priority via the transport network. Misconfiguration of priority settings may therefore cause some delays for the cell broadcast delivery in case of network congestion but will unlikely cause unavailability of cell broadcast. Therefore, the probability that misconfiguration causes unavailability of cell broadcast is rated as LOW .	Low
Insufficient Access Controls	Availability (AP2-A)	Adversaries may disable routers located at cell sites or in data centers, e.g. via physical or cyber-attack, which may impair or disable cell broadcast, e.g. in conjunction with a terror attack. Given that such a coordinated attack would require special resources, skills, and sophistication, it is rated as having LOW probability.	Low
Implementation Flaw (Quality of the Product)	Availability (AP3-A)	Implementation flaws are considered as having MEDIUM probability to cause unavailability of cell broadcast, as such flaws may have a variety of causes and may be non-trivial to detect, in particular due to the rare occurrence of cell broadcast. For example, SW upgrades can lead to the use of flawed or misconfigured SW. Faulty SW updates were the single largest cause of telecom user hours lost in the EU in the year 2022 [38].	Medium
Specification Weakness	Availability (AP4-A)	The probability of unavailability of cell broadcast due to specification weaknesses in the 5G transport network is rated as LOW as the 5G transport network incorporates state-of-art transport protocols with robust QoS mechanisms defined by IETF.	Low

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Failure of Power Supply	Availability (AP5-A)	Unavailability of cell broadcast may be caused by a (multi-) regional power outage or by a power outage affecting a data center [64], [74], [80] hosting transport network equipment. Such a power outage may be caused by human error, power system failure, a natural disaster [8], [11], a terror attack, ageing infrastructure or theft of components [30]. Given the variety of different sources for failures of power supply, their probability to be the cause of unavailability of a cell broadcast is rated as MEDIUM . For example, in 2021, a massive power outage in Europe affected several countries, including, France and Italy. The power outage was caused by a grid disconnection that led to significant disruptions to telecommunications services, including the unavailability of cell broadcast systems used for emergency alerts [30].	Medium
Failure of Hardware Components or Unavailability of Spare Parts	Availability (AP6-A)	The probability of a hardware failure to cause unavailability of cell broadcast is rated as MEDIUM , as there are a wide variety of components that may cause the failure, such as failed routing or switching equipment, optical or microwave transport equipment, optic link failure [68], physical damage of connectors, cabinets, or microwave masts, etc., and a variety of failure sources such as human error, a natural disaster [8], [11], a terror attack, or sabotage or theft of components [30], [75]. HW failures were the single largest root cause of telecom security incidents in the EU in the year 2021 [38].	Medium

5.2.3 Telecommunications Secrecy (TTDSG § 3)

Telecommunication secrecy, ensuring the confidentiality and privacy of communications, is an important aspect of 5G networks. These are some examples for relevant threats:

- *Eavesdropping*: Unauthorized interception of sensitive data during transmission is a significant threat. The probability of eavesdropping depends on the effectiveness of encryption and security measures in place. It is generally low if encryption is implemented.
- *Network Intrusion*: Unauthorized access to network components, such as routers or switches, can lead to the interception of traffic or unauthorized monitoring. This risk's probability depends on the network's security measures, patch management, and access control. Properly secured networks have a lower probability of intrusion.
- *Physical Security Breaches*: Unauthorized access to network equipment or facilities can compromise data secrecy. The likelihood of physical security breaches depends on the physical security measures in place. Proper fencing, surveillance, and access controls can minimize this risk.
- *Data Leaks*: Inadequate handling of sensitive data within the network, including logs and databases, can lead to information leakage. The probability of data leaks depends on network management practices and data security measures. Proper data handling and access control are essential to mitigate this risk. [10]

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Misconfiguration	Confidentiality (AP1-C)	The probability of telecommunications secrecy issues due to misconfigurations is rated as LOW , as there are a limited number of security parameters that can be configured by the transport network management, and in many cases, encryption is additionally applied on application layer or on RAN or core network interfaces. For example, the likelihood of a misconfiguration related to the usage of deprecated cipher suites in the TLS protocol is low, given frequent updates and proactive security measures implemented in modern systems guaranteeing data confidentiality [85].	Low
Insufficient Access Controls	Confidentiality (AP2-C)	Eavesdropping would require physical access to the transport medium, e.g. the fiber optical cable, and disabling the encryption between transport network elements, e.g., by means of a cyber-attack [29]. Such attack would require plenty of resources and sophistication beyond insufficient access control. Therefore, the probability of confidentiality issues with telecommunications secrecy due to insufficient access controls is rated as LOW .	Low
Implementation Flaw (Quality of the Product)	Confidentiality (AP3-C)	Implementation flaws are considered as having MEDIUM probability to cause telecommunications secrecy issues, as such flaws may have a variety of causes and may be non-trivial to detect, for example: • Usage of outdated transport protocols with known security issues such as TLS 1.1 or older [40], [70], [81]. • Flawed transport network security architecture, e.g. not deploying a proper security key distribution system. • Faulty implementations that leak the secret key used for encryption, e.g. by side-channel attacks such as power analysis or time inconsistency attacks. • Deployment of fiber optical transport equipment lacking intrusion detection capabilities.	Medium
Specification Weakness	Confidentiality (AP4-C)	The probability of telecommunications secrecy issues due to specification weaknesses in the 5G transport network is rated as LOW as the 5G system incorporates secure, state-of-art transport protocols defined by IETF such as IP layer security (IPsec) or Transport Layer Security (TLS) [84], [85].	Low
Failure of Power Supply	Confidentiality (AP5-C)	The failure of power supply appears unrelated to telecommunications secrecy.	Low
Failure of Hardware Components or Unavailability of Spare Parts	Confidentiality (AP6-C)	The failure of hardware components or unavailability of spare parts appears unrelated to telecommunications secrecy.	Low

5.2.4 Emergency Preparedness (TKG § 184)

Emergency preparedness in the context of 5G networks is crucial to ensure that communication services remain available and reliable during times of crises, such as natural disasters.

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Misconfiguration	Availability (AP1-A)	Emergency preparedness requires high availability of the transport network and consistent prioritization of traffic related to high priority users such as police or first responder teams across the transport network. The transport network typically includes many network elements on the route from the cell site to the emergency call center that may be negatively impacted by misconfiguration. On the other hand, the transport network capacity is often generously dimensioned and able to deal with traffic peaks, particularly if originating from a small number of radio cells. Therefore, the probability that misconfiguration disables emergency preparedness is rated as low to MEDIUM .	Medium
Insufficient Access Controls	Availability (AP2-A)	Adversaries may disable aggregation routers deployed at the radio cell sites or in a data center e.g. in conjunction with a terror or military attack to compromise work of police or first responders. As cyber-attacks [36], [69] and physical attacks [35], [49] on technical infrastructure have become more widespread the probability of impaired or disabled emergency preparedness being caused by insufficient access controls is rated as MEDIUM .	Medium
Implementation Flaw (Quality of the Product)	Availability (AP3-A)	Implementation flaws are considered as having MEDIUM probability to impair or disable emergency preparedness, as such flaws may have a variety of causes and may be non-trivial to detect, for example: - Flaws in the implementation of traffic prioritization may remain undetected due to the rare occurrence of emergency situations. - SW upgrades can lead to the use of flawed or misconfigured SW [1], [66]. - Flawed architecture of the optical transport network may enable alien wavelength attacks [78] impairing network performance.	Medium
Specification Weakness	Availability (AP4-A)	The probability of impaired or disabled prioritization in emergency situations due to specification weaknesses in the 5G transport network is rated as LOW as the 5G transport network incorporates state-of-art transport protocols with robust QoS mechanisms defined by IETF.	Low
Failure of Power Supply	Availability (AP5-A)	Natural disasters such as flooding, fire or earthquakes are relevant causes for emergency scenarios [11], [30]. As such disaster events can simultaneously cause power outages [25], possibly over larger geographical areas, the probability of occurrence that emergency preparedness is impaired or disabled by a failure of power supply is rated as MEDIUM . Further, also adversaries may disable power supply, e.g. in conjunction with a terror attack to compromise the work of police or first responder teams [30], [75]. The arson attack on energy infrastructure at Grünheide may serve as an example [35].	Medium

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Failure of Hardware Components or Unavailability of Spare Parts	Availability (AP6-A)	Natural disasters such as flooding, fire or earthquakes are relevant causes for emergency scenarios [8], [11]. As such disaster events can simultaneously cause hardware failures, possibly over larger geographical areas, the probability of occurrence that emergency preparedness is impaired or disabled by a hardware failure is rated as MEDIUM .	Medium

5.2.5 Regional Network Failure

Regional Network Failure refers to a situation where a significant portion of a network within a specific geographic region experiences a disruption or outage, impacting the connectivity and services for users in that area. [2]

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Misconfiguration	Availability (AP1-A)	A regional network failure may be caused by misconfiguration via the central transport network management system(s) controlling a variety of transport protocols and many network elements such as routers or switches, or optical or microwave transmission equipment. The transmission route may be comprised of different technologies, e.g. a mix of optical and microwave transmission, possibly provided by different equipment vendors and/or using different network management systems. Misconfigured routers and switches with incorrect timing can lead to inefficient management of data traffic, resulting in delays and lost packets. Given that multiple configuration domains with a wide range of parameters and multiple responsible individuals can be involved, the probability of a misconfiguration causing a regional network failure is rated as MEDIUM .	Medium
Insufficient Access Controls	Availability (AP2-A)	An adversary may cause a regional network failure by gaining access to the central network configuration management system(s), e.g. by means of a cyber-attack, or by physically attacking a regional data center hosting transport network elements used in the telecom network. As cyber-attacks [36], [69] and physical attacks [35], [49] on technical infrastructure have become more widespread the probability of a regional network failure being caused by insufficient access controls is rated as MEDIUM .	Medium

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Implementation Flaw (Quality of the Product)	Availability (AP3-A)	Implementation flaws are considered as having MEDIUM probability to cause regional network failures, as such flaws may have a variety of causes and may be non-trivial to detect, for example: • SW upgrades can lead to the use of flawed or misconfigured SW [1], [7], [66]. • Transport equipment provided by multiple vendors or use of a mix of transmission technologies (incl. legacy technologies such as SONET/SDH) may cause interoperability issues, particularly in conjunction with SW updates. • Flawed dimensioning with respect to link capacity or route redundancy may cause network congestion. • Flawed architecture of the optical transport network may enable alien wavelength attacks [78] impairing network performance.	Medium
Specification Weakness	Availability (AP4-A)	The probability of a regional network failure due to specification weaknesses in the 5G transport network is rated as LOW as the 5G transport network incorporates state-of-art transport protocols with robust QoS mechanisms defined by IETF.	Low
Failure of Power Supply	Availability (AP5-A)	A regional network failure can be caused by a regional power outage or by a power outage in one or several data centers hosting transport network equipment [73]. Such power outages can be caused by human error, a power grid failure, a natural disaster [8], [13] or theft of components [11]. In view of the numerous causes of power outages, the probability that these will lead to a regional power outage is classified as MEDIUM .	Medium
Failure of Hardware Components or Unavailability of Spare Parts	Availability (AP6-A)	A hardware failure leading to a regional network outage would need to occur at a more central location in the network away from the cell sites. As the routed/switched transport connections provide redundant hardware resources and transmission paths, the probability of a hardware failure causing a failure of the regional network is rated as LOW .	Low

5.2.6 Multi-Regional Network Failure

A multi-regional network failure in a 5G transport network refers to a disruption that affects connectivity and services across several geographic regions. This failure occurs when the network experiences severe issues or collapses in multiple areas simultaneously, impacting communication and data transmission on a large scale. Overall, a multi-regional network failure in a 5G transport network represents a significant challenge for service providers, users, and emergency responders due to its wide-reaching impact, requiring swift and effective measures to mitigate its effects and restore network functionality [5].

A multi-regional transport network can be assumed to include several regional transport networks, e.g. used to connect the 5G radio cell sites to the respective regional data centers, and wide area transport networks connecting the regional data centers to centralized network locations, e.g. hosting the core network functions and/or providing access to the internet.

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Misconfiguration	Availability (AP1-A)	A multi-regional network failure may be caused by misconfiguration via the central transport network management system(s). Compared to a regional network failure, a multi-regional network failure can be expected to occur less likely as a result of misconfiguration for the following reasons: - The multi-regional network may be comprised of regional subnetworks, such that the impact of a misconfiguration may be confined to the regional subnetwork. - The multi-regional network can be expected to have ample of redundant transport routes. On the other hand, the multi-regional network comprises a larger number of network components, including wide-area transport components, not present in the regional transport network(s). From the above, the probability of a misconfiguration causing a multi-regional network failure is rated as low to MEDIUM.	Medium
Insufficient Access Controls	Availability (AP2-A)	An adversary may cause a multi-regional network failure by gaining access to the central network configuration management system(s), e.g. by intruding into and misconfiguring the wide area transport nodes by means of a cyber-attack. To cause a multi-regional network failure by a physical attack requires very high effort, due to inherent redundancy in the wide area network, e.g. cutting a fiber optical cable is assumed to likely not easily cause a multi-regional network failure. As cyber-attacks (and physical attacks) [36], [69], [35], [49] on technical infrastructure have become more widespread the probability of a regional network failure being caused by insufficient access controls is rated as MEDIUM.	Medium
Implementation Flaw (Quality of the Product)	Availability (AP3-A)	Implementation flaws are considered as having MEDIUM probability to cause multi-regional network failures, as such flaws may have a variety of causes and may be non-trivial to detect, for example: - SW upgrades can lead to the use of flawed or misconfigured SW [1], [7], [66]. - Transport equipment provided by multiple vendors or use of a mix of transmission technologies may cause interoperability issues, particularly in conjunction with SW updates or if multiple network operators are involved on the transport route. - Flawed dimensioning with respect to link capacity or route redundancy may cause network congestion. - Flawed architecture of the optical transport network may enable alien wavelength attacks [78] impairing network performance.	Medium
Specification Weakness	Availability (AP4-A)	The probability of a multi-regional network failure due to specification weaknesses in the 5G transport network is rated as LOW as the 5G transport network incorporates state-of-art transport protocols with robust QoS mechanisms defined by IETF.	Low
Failure of Power Supply	Availability (AP5-A)	Natural disasters such as floods, fires or earthquakes are relevant causes for multi-regional network failure [8], [11] [85]. As such disaster events can simultaneously cause power outages [12], [13], possibly over larger geographical areas, the probability of occurrence that multi-regional network failure will be caused by a failure of power supply is rated as MEDIUM. In addition, attackers may disable power supply, e.g. in conjunction with a terror attack to compromise the transport network.	Medium

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Failure of Hardware Components or Unavailability of Spare Parts	Availability (AP6-A)	A hardware failure leading to a multi-regional network outage would need to occur at a more central location in the network away from the cell sites. As the routed/switched transport connections provide redundant hardware resources and transmission paths, the probability of a hardware failure causing a failure of the multi-regional network is rated as LOW .	Low

5.2.7 Failure of Mobile Service

The failure of mobile service is considered a relevant attack domain due to its significant impact on both individuals and businesses. [9]

For the transport network to cause failure of mobile service implies outage of the transport network over a large geographical area. For the failure of mobile service scenario, therefore the argumentation related with multi-regional network failure (Section 5.2.6) applies.

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Misconfiguration	Availability (AP1-A)	A multi-regional network failure may be caused by misconfiguration via the central transport network management system(s). Compared to a regional network failure, a multi-regional network failure can be expected to occur less likely as a result of misconfiguration for the following reasons: - The multi-regional network may be comprised of regional subnetworks, such that the impact of a misconfiguration may be confined to the regional subnetwork. - The multi-regional network can be expected to have ample of redundant transport routes. On the other hand, the multi-regional network comprises a larger number of network components, including wide-area transport components, not present in the regional transport network(s). From the above, the probability of a misconfiguration causing a multi-regional network failure is rated as low to MEDIUM.	Medium
Insufficient Access Controls	Availability (AP2-A)	An adversary may cause a multi-regional network failure by gaining access to the central network configuration management system(s), e.g. by intruding into and misconfiguring the wide area transport nodes by means of a cyber-attack. To cause a multi-regional network failure by a physical attack would likely require very high effort, due to inherent redundancy in the wide area network, e.g. cutting a fiber optical cable is assumed to likely not easily cause a multi-regional network failure. As cyber-attacks (and physical attacks) [36], [69], [35], [49] on technical infrastructure have become more widespread the probability of a regional network failure being caused by insufficient access controls is rated as MEDIUM.	Medium

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Implementation Flaw (Quality of the Product)	Availability (AP3-A)	Implementation flaws are considered as having MEDIUM probability to cause multi-regional network failures, as such flaws may have a variety of causes and may be non-trivial to detect, for example: - SW upgrades can lead to the use of flawed or misconfigured SW [1], [7], [66]. - Transport equipment provided by multiple vendors or use of a mix of transmission technologies may cause interoperability issues, particularly in conjunction with SW updates or if multiple network operators are involved on the transport route. - Flawed dimensioning with respect to link capacity or route redundancy may cause network congestion. Flawed architecture of the optical transport network may enable alien wavelength attacks [78] impairing network performance.	Medium
Specification Weakness	Availability (AP4-A)	The probability of failure of mobile service due to specification weaknesses in the 5G transport network is rated as LOW as the 5G transport network incorporates state-of-art transport protocols with robust QoS mechanisms defined by IETF.	Low
Failure of Power Supply	Availability (AP5-A)	Natural disasters such as floods, fires or earthquakes are relevant causes for multi-regional network failure [8], [11] [85]. As such disaster events can simultaneously cause power outages [12], [13], possibly over larger geographical areas, the probability of occurrence that multi-regional network failure will be caused by a failure of power supply is rated as MEDIUM . In addition, attackers may disable power supply, e.g. in conjunction with a terror attack to compromise the transport network.	Medium
Failure of Hardware Components or Unavailability of Spare Parts	Availability (AP6-A)	A hardware failure leading to a multi-regional network outage would need to occur at a more central location in the network away from the cell sites. As the routed/switched transport connections provide redundant hardware resources and transmission paths, the probability of a hardware failure causing a failure of the multi-regional network is rated as LOW .	Low

5.2.8 Fraud

Fraud is not directly tied to the 5G transport network. However, it poses indirect risks. Unauthorized access attempts could compromise relevant network components, impacting security.

Similarly, billing fraud, while service-centric, could indirectly impact infrastructure investment and maintenance. Security breaches arising from fraud could compromise transport network elements, potentially affecting data transport. Addressing fraud aligns with broader network security strategies, ensuring overall integrity and security.

5.2.9 Subscriber-Data Leak

Data leaks can be highly relevant in the context of 5G backhaul and backbone networks. These networks handle substantial volumes of sensitive and vital data, making data leaks a significant concern due to the potential repercussions.

5G backhaul and backbone networks transmit a vast array of data, including user information, corporate data, and confidential communications. A leak in this data can compromise user privacy, corporate secrets, or even infrastructure data. Data transmitted through 5G networks often falls under various regulations and compliance standards, such as data protection laws (GDPR, CCPA). Any data leak could lead to non-compliance, resulting in legal consequences and financial penalties for the network operators.

In some cases, a data leak could be part of a larger cyberattack aiming to disrupt services. It might involve overwhelming the network with data traffic or targeting specific nodes to interrupt service availability [10].

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Misconfiguration	Confidentiality (AP1-C)	Incorrect configuration settings, such as improper routing or firewall settings, can lead to data leaks. Improper routing protocol configurations in the Border Gateway Protocol (BGP) can result in incorrect route advertisements, potentially exposing internal network routers to external entities [71]. These misconfigurations are often the result of human error or inadequate security protocols. Nevertheless, the transmitted data is typically encrypted in other layers of the network (e.g. on application layer or on the 5G network interfaces), resulting in a LOW probability of unencrypted data leak due to misconfigurations.	Low
Insufficient Access Controls	Confidentiality (AP2-C)	Lack of proper access controls can allow unauthorized access to sensitive data in the transport network, leading to data leaks. Insufficient access controls often involve weak authentication mechanisms or excessive permissions, allowing an attacker to access sensitive data stored in administrative panels of routers and switches, e.g. routing tables, IP addresses and DHCP leases, access control lists and security certificates. However, as actual user data from telecommunication connections is not stored but only forwarded from router to router, the probability of a data leak coming from insufficient access control is LOW .	Low
Implementation Flaw (Quality of the Product)	Confidentiality (AP3-C)	Flaws in software or hardware implementations, such as bugs or design errors, can create vulnerabilities that lead to data leaks. For example, buffer overflow vulnerabilities, exploited via specially crafted packets sent to routers, can expose sensitive data to the attacker. Furthermore, insecure firmware updates for routers, switches and bridges can be compromised and leak data to the attacker. These flaws often result from inadequate testing and quality assurance processes. Due to the broad attack surface in combination with the threat of unpatched vulnerabilities, the probability for implementation flaws causing data leaks is estimated as MEDIUM .	Medium
Specification Weakness	Confidentiality (AP4-C)	Inadequate specifications during the design phase can lead to weaknesses in data handling and transmission, resulting in potential data leaks. These weaknesses often stem from incomplete or incorrect requirements. However, the specifications are a product of high-quality research and intensive elaboration, leading to a LOW probability for any specification weaknesses to cause a data leak.	Low

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Failure of Power Supply	Confidentiality (AP5-C)	Power supply issues of transport network equipment generally do not directly cause data leaks but can lead to system outages that expose vulnerabilities during recovery or restart processes. These power failures might indirectly result in data leaks due to improper shutdowns or system crashes. Nevertheless, data leaks from sudden power outages are uncommon [25]. Therefore, the power failure does not appear to be related to the confidentiality of the data leakage and is rated as LOW or unrelated.	Low
Failure of Hardware Components or Unavailability of Spare Parts	Confidentiality (AP6-C)	Failure of critical hardware components, such as routers or switches, or the unavailability of spare parts can lead to data exposure during replacement or repair processes. These failures can also cause system downtimes that might be exploited by attackers. However, inside the transport network, data leaks are rarely caused by hardware failures. Therefore, the hardware failures do not appear to be related to the confidentiality of the data leakage and is rated as LOW or unrelated.	Low

5.3 User Perspective

This section aims to assess the likelihood of scenarios by examining potential attack paths within the transport network. It focuses on risks from a public network user's perspective and doesn't delve into considerations specific to private 5G networks. Additionally, it avoids reiterating scenarios covered earlier, such as those concerning emergency calls.

5.3.1 Urgent Telephone Calls Not Possible

The risk scenario of Urgent Telephone Calls Not Possible is typically associated with the availability and reliability of communication networks. If network components, such as routers, switches, and communication servers, are designed with high reliability and redundancy, the probability of urgent telephone calls becoming impossible due to network failures would be low. The effectiveness of security measures, including encryption, authentication, and intrusion detection, plays a role. If security is robust, the probability of malicious activities disrupting urgent calls decreases.

Misconfigurations in essential network elements could impact the availability of services. If there are rigorous configuration management practices, the probability of configuration-related issues affecting urgent calls is lower.

The efficiency of the response to network incidents is crucial. If there are rapid response and mitigation procedures in place, the probability of prolonged disruptions to urgent calls decreases. However, it is essential to regularly assess and enhance network resilience, security measures, and incident response capabilities to maintain a low probability of such disruptions.

The risk scenario of urgent telephone calls not possible may be related with a regional or other type of network failure, e.g. a localized network failure may disable users to make an urgent phone call, particularly indoor users able to receive one or a few cells of the network only.

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Misconfiguration	Availability (AP1-A)	A misconfiguration via the central transport network management system(s) controlling a variety of transport protocols [20] and many network elements such as routers or switches, or optical or microwave transmission equipment can lead to urgent telephone calls not being possible. The transmission route may be comprised of different technologies, e.g. a mix of optical and microwave transmission, possibly provided by different equipment vendors and/or using different network management systems. Given that multiple configuration domains with a wide range of parameters and multiple responsible individuals can be involved, the probability of a misconfiguration causing urgent telephone calls not possible is rated as MEDIUM .	Medium
Insufficient Access Controls	Availability (AP2-A)	An adversary may cause urgent telephone calls not possible by gaining access to the central network configuration management system(s), e.g. by means of a cyber-attack, or by physically attacking a regional data center hosting transport network elements used in the telecom network. As cyber-attacks [36], [69] and physical attacks [35], [49] on technical infrastructure have become more widespread the probability of urgent telephone calls not being possible due to insufficient access controls is rated as MEDIUM .	Medium
Implementation Flaw (Quality of the Product)	Availability (AP3-A)	Implementation flaws are considered as having MEDIUM probability to cause urgent telephone calls not possible, as such flaws may have a variety of causes and may be non-trivial to detect, for example: • SW upgrades may result in the use of flawed or misconfigured SW [1], [7], [66]. • Multi-vendor transport equipment or use of a mix of transmission technologies (incl. legacy technologies such as SONET/SDH) can lead to interoperability issues, especially in conjunction with SW updates. • Flawed dimensioning with respect to link capacity or route redundancy may cause network congestion. • Flawed architecture of the optical transport network may allow alien wavelength attacks [78] that degrade network performance.	Medium
Specification Weakness	Availability (AP4-A)	The probability of urgent telephone calls not possible due to specification weaknesses in the 5G transport network is rated as LOW as the 5G transport network incorporates state-of-art transport protocols with robust QoS mechanisms defined by IETF.	Low
Failure of Power Supply	Availability (AP5-A)	A localized or regional power outage can make urgent phone calls impossible [42]. Such power outages can be caused by human error, power system failures, natural disasters [8], [11] terrorist attacks, sabotage or theft of critical components, e.g. [49]. Given the diverse range of potential causes for power supply disruptions, the probability of these outages leading to urgent telephone calls not being possible is rated as MEDIUM .	Medium
Failure of Hardware Components or Unavailability of Spare Parts	Availability (AP6-A)	The probability of a hardware failure to cause a localized network failure leading to urgent telephone calls not possible is rated as MEDIUM , as there are a wide variety of components that may cause the failure, such as failed routing or switching equipment deployed at the radio sites in or data centers, cable breaks [68], physical damage of connectors, cabinets, etc., and a variety of failure sources such as human error, a natural disaster [8], [11] a terror attack, e.g. [35], or sabotage or theft of components, e.g. [30], [75]. HW failures were the single largest root cause of telecom security incidents in the EU in the year 2021 [38].	Medium

5.3.2 Authenticity of the Receiver of a Call

The authenticity of the receiver in a call generally pertains to authentication mechanisms at higher network layers, often within application or communication protocols, rather than the 5G transport network specifically. In the context of telecommunications and calls, ensuring the authenticity of the receiver involves protocols and mechanisms related to call signaling, authentication of users/devices, and authorization procedures implemented in the application layers or within telecommunication protocols like SIP (Session Initiation Protocol) or other signaling protocols.

However, in the context of 5G transport networks, these considerations are more focused on the functionalities of the hardware and network components rather than the specific intricacies of receiver authentication.

5.3.3 Integrity of Transmitted Data

Ensuring the integrity of transmitted data is a vital aspect of communication security. In the context of the discussed scenarios, risks related to the integrity of transmitted data could be associated with vulnerabilities in encryption, integrity protection, or potential man-in-the-middle attacks.

For example, an implementation flaw in the integrity protection of a DWDM system compromises the integrity of transmitted data.

Considering encryption strength, integrity protection mechanism, security of communication protocols, and configuration management the probability of issues related to the integrity of transmitted data can be estimated as generally low to medium. Regular security assessments, adherence to best practices in encryption and integrity protection, and prompt updates to address vulnerabilities contribute to maintaining a high level of data integrity within the network.

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Misconfiguration	Integrity (AP1-I)	Misconfigurations in transport network hardware (e.g. routers, switches, etc.) are a frequent cause for cyberattacks. Attacks on the Border Gateway Protocol (BGP) in case it is misconfigured allow an attacker to announce false routers, redirect traffic and thus intercept and alter it [71]. Similarly, a misconfigured Network Time Protocol (NTP) can be exploited to manipulate time synchronization [14] across devices, allowing potential replay attacks. In this case, the integrity of transmitted data cannot be guaranteed. Exploiting misconfigured protocols for integrity breaches is a common attack, resulting in a MEDIUM probability for this attack path category.	Medium
Insufficient Access Controls	Integrity (AP2-I)	Unauthorized intrusions often stem from weak authentication mechanisms and poor access control, e.g. for administrator interfaces or publicly accessible ports on a router. In most cases, these control mechanisms are implemented by network administrators, which may lead to unintended security vulnerabilities. Data integrity is at risk if an attacker manages to gain unintended access rights to transport network equipment. In this case, man-in-the-middle attacks are possible, allowing an attacker to intercept and alter messages sent by users. In general, the probability of an intruder tampering with user data, once he gained full access to a network component, is rated as MEDIUM .	Medium

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Implementation Flaw (Quality of the Product)	Integrity (AP3-I)	Erroneous implementations of cryptographic ciphers or the usage of legacy ciphers can directly affect the authenticity of messages, e.g. by intercepting, decrypting, modifying and re-encrypting the data. In addition, reused signature keys can be recovered and used to re-sign forged messages [29]. Such implementation weaknesses can be exploited by attackers to alter transmitted messages without the users realizing. Software-side attacks are a common cause for cyberattacks, especially in case of data integrity, as these implementation flaws can enable man-in-the-middle attacks, which allow the attacker to tamper with transmitted data. Hence, implementation flaws are rated as having a MEDIUM probability to occur and breach data integrity.	Medium
Specification Weakness	Integrity (AP4-I)	Specification weaknesses can enable label spoofing attack, in which attackers can impersonate other network devices and thus breach the integrity. However, these kinds of attacks have a rather LOW probability, as specifications typically represent robust and sophisticated guidelines on network architectures. Protocols used in transport networks, such as multiprotocol label switching (MPLS) are designed to handle errors and reroute traffic in case of anomalies.	Low
Failure of Power Supply	Integrity (AP5-I)	Failure of power supply appears unrelated to the integrity of transmitted data.	Low
Failure of Hardware Components or Unavailability of Spare Parts	Integrity (AP6-I)	Failure of hardware components appears unrelated to the integrity of transmitted data.	Low

5.3.4 User's Identity

Ensuring the authenticity of a user's identity is paramount in any network, including the 5G Backhaul and Backbone Network. Risks related to the authenticity of a user's identity might involve vulnerabilities in authentication processes, potential identity theft, or weaknesses in user verification mechanisms.

The authenticity of a user's identity within the 5G Backhaul and Backbone Network is contingent on the effectiveness of authentication mechanisms, the security of identity databases, and the overall strength of user verification processes.

Tapping and misusing the identity of a user itself might not be considered a typical risk scenario within the 5G transport network, especially in the context of hardware or network components. Instead, user identity management is a crucial aspect of network security but is more pertinent to access control and authentication within the network's operational aspects rather than being a risk scenario in the transport network.

5.3.5 Eavesdropping of a Communication

Eavesdropping on communication is a significant security concern, and in the context of the 5G Backhaul and Backbone Network, this risk typically involves unauthorized interception of transmitted data. Potential scenarios contributing to the risk of eavesdropping could include vulnerabilities in encryption mechanisms, weaknesses in network security, or exploitation of communication protocols.

The probability of eavesdropping on communication within the 5G Backhaul and Backbone Network depends on the strength of encryption, the effectiveness of authentication mechanisms, and the overall security posture.

Considering these factors, the probability of eavesdropping on communication within the 5G Backhaul and Backbone Network can be estimated as generally low to medium. Regular security assessments, adherence to best practices in encryption and authentication, and proactive measures against potential vulnerabilities contribute to maintaining a high level of communication confidentiality.

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Misconfiguration	Confidentiality (AP1-C)	An attacker can eavesdrop a communication if cryptographic parameters for data encryption are not chosen thoughtfully [29]. For example, default encryption might be turned off or an outdated TLS version (SSL) might be configured as a supported and available option [81]. In addition, misconfigured ports can allow attackers to connect to these open ports and capture all traffic, although the data is typically encrypted on further layers of the network. Therefore, the probability of network misconfigurations being used for communication eavesdropping and the interception of unencrypted data is rated as LOW .	Low
Insufficient Access Controls	Confidentiality (AP2-C)	Access control usually prohibits attackers from intruding a network and intercepting messages. If access control is not defined strictly enough, it may offer possibilities for an attacker to enter a network and eavesdrop the communication. Additionally, physical access to transport network equipment, i.e. optic cables, allows an attacker to inject their device, monitor the traffic and capture sensitive information. However, due to encryption on the application layer or encryption on the network interfaces of the 5G system, the interception of encrypted data does not reveal confident information to the attacker. Therefore, the probability for communication eavesdropping in a decrypted form is rated as LOW .	Low
Implementation Flaw (Quality of the Product)	Confidentiality (AP3-C)	Software implementation of protocols may contain vulnerabilities that allow attackers to invade a network and eavesdrop a communication. For example, faulty implementations of encryption algorithms or key distribution mechanisms can allow attackers to perform side-channel attacks such as power analysis or time correlation attacks in order to extract the secret key and breach the confidentiality of transmitted data. In addition, vulnerabilities in the firmware of routers, switches and other optical network equipment may enable an attacker to execute malicious code and inject commands to eventually monitor incoming and outgoing data. If these vulnerabilities are not fixed regularly, attackers can exploit them and eavesdrop a communication. The probability of an implementation flaw causing the disclosure of confidential transmitted data is rated as MEDIUM .	Medium

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Specification Weakness	Confidentiality (AP4-C)	The probability for specification exploits regarding eavesdropping of communication data is LOW if state-of-art encryption and authentication algorithms are used. Generally, modern 5G architectures and implementations use the most recent algorithms and thus reduce the risk for being breached [84], [85].	Low
Failure of Power Supply	Confidentiality (AP5-C)	Failure of power supply appears unrelated to the confidentiality of a communication.	Low
Failure of Hardware Components or Unavailability of Spare Parts	Confidentiality (AP6-C)	Failure of hardware components appears unrelated to the confidentiality of a communication.	Low

5.3.6 Slowing Down/Restricting the Connection

Slowing down or restricting the connection, often referred to as network throttling or intentional bandwidth limitation, can be a deliberate action by network administrators or, in some cases, a result of malicious activities. This risk in the 5G Backhaul and Backbone Network could be associated with various factors such as network congestion, misconfigurations, or intentional interference.

The probability of this risk scenario within the 5G Backhaul and Backbone Network can be influenced by various factors, including network design, traffic management practices, and potential malicious activities

- If the network is well-designed with sufficient capacity to handle the expected traffic, the probability of unintentional slowdowns or restrictions is lower.
- The effectiveness of Quality of Service (QoS) settings and traffic management practices on routers and switches is crucial. Properly configured QoS can minimize the probability of specific connections being inadvertently slowed down or restricted.
- Misconfigurations in network devices, particularly those related to traffic prioritization or bandwidth allocation, may impact connection speeds. Rigorous configuration management practices reduce the probability of misconfigurations affecting connection performance.
- Malicious activities, such as Denial-of-Service (DoS) attacks, can intentionally slow down or restrict network connections. The probability of intentional interference depends on the security measures in place to mitigate such attacks.

Considering these factors, the probability of slowing down or restricting the connection within the 5G Backhaul and Backbone Network can be estimated as generally low to medium. Regular network monitoring, effective traffic management, and robust security practices contribute to maintaining a high level of connection performance and availability.

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Misconfiguration	Availability (AP1-A)	A misconfiguration via the central transport network management system(s) controlling a variety of transport protocols and many network elements such as routers or switches, or optical or microwave transmission equipment can lead to slowing down or restricting the connection. The transmission route may be comprised of different technologies, e.g. a mix of optical and microwave transmission, possibly provided by different equipment vendors and/or using different network management systems. Given that multiple configuration domains with a wide range of parameters and multiple responsible individuals can be involved, the probability of a misconfiguration causing slowing down or restricting the connection is rated as MEDIUM .	Medium
Insufficient Access Controls	Availability (AP2-A)	An adversary may cause connection failure by gaining access to the central network configuration management system(s), e.g. by means of a cyber-attack, or by physically attacking a regional data center hosting transport network elements used in the telecom network. As cyber-attacks and physical attacks on technical infrastructure have become more widespread the probability of slowing down or restricting the connection due to insufficient access controls is rated as MEDIUM .	Medium
Implementation Flaw (Quality of the Product)	Availability (AP3-A)	Implementation flaws are considered as having MEDIUM probability to cause slowdown or restriction of connection, as such flaws may have a variety of causes and may be non-trivial to detect, for example: • SW upgrades may result in the use of flawed or misconfigured SW [1], [7], [66]. • Multi-vendor transport equipment or use of a mix of transmission technologies (incl. legacy technologies such as SONET/SDH) can lead to interoperability issues, especially in conjunction with SW updates. • Flawed dimensioning with respect to link capacity or route redundancy may cause network congestion. • Flawed architecture of the optical transport network may allow alien wavelength attacks [78] that degrade network performance.	Medium
Specification Weakness	Availability (AP4-A)	The probability of slowing down or restricting the connection due to specification weaknesses in the 5G transport network is rated as LOW as the 5G transport network incorporates state-of-art transport protocols with robust QoS mechanisms defined by IETF.	Low
Failure of Power Supply	Availability (AP5-A)	A localized or regional power outage can slow down or restrict the connection [42]. Such power outages can be caused by human error, power system failures, natural disasters, terrorist attacks, sabotage or theft of critical components, e.g. [49]. Given the many possible causes of power outages, the probability that these outages will lead to a slowdown or restriction of connection is rated as MEDIUM .	Medium
Failure of Hardware Components or Unavailability of Spare Parts	Availability (AP6-A)	The probability of a hardware failure to slow down or restrict the connection is rated as MEDIUM , as there are a wide variety of components that may cause the failure, such as failed routing or switching equipment deployed at the radio sites in or data centers, cable breaks, physical damage of connectors, cabinets, etc., and a variety of failure sources such as human error, a natural disaster [8], a terror attack, e.g. [35], [30], [75] or sabotage or theft of components, e.g. [49]. HW failures were the single largest root cause of telecom security incidents in the EU in the year 2021 [38].	Medium

5.3.7 Connection Failure

Connection failure in the 5G Backhaul and Backbone Network can result from various factors, including hardware failures, misconfigurations, or external attacks.

If essential network components such as routers, switches, and servers are designed with high reliability, redundancy, and failover mechanisms, the probability of connection failures is generally lower.

Misconfigurations in network devices, whether related to routing tables, security settings, or other parameters, can impact the stability of connections. Rigorous configuration management practices reduce the probability of misconfigurations leading to connection failures.

The efficiency of response mechanisms to hardware failures, such as prompt identification and replacement of faulty components, influences the probability of prolonged connection failures.

The presence of security measures, including intrusion detection and prevention systems, helps mitigate the risk of connection failure due to intentional attacks or malicious activities.

Considering these factors, the probability of connection failure in the 5G Backhaul and Backbone Network can be estimated as generally medium. Regular maintenance, proactive monitoring, and adherence to best practices in network design and security contribute to maintaining a high level of connection reliability.

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Misconfiguration	Availability (AP1-A)	A misconfiguration via the central transport network management system(s) controlling a variety of transport protocols and many network elements such as routers or switches, or optical or microwave transmission equipment can lead to connection failures. The transmission route may be comprised of different technologies, e.g. a mix of optical and microwave transmission, possibly provided by different equipment vendors and/or using different network management systems. Given that multiple configuration domains with a wide range of parameters and multiple responsible individuals can be involved, the probability of a misconfiguration causing connection failure is rated as MEDIUM .	Medium
Insufficient Access Controls	Availability (AP2-A)	An adversary may cause connection failures by gaining access to the central network configuration management system(s), e.g. by means of a cyber-attack, or by physically attacking a regional data center hosting transport network elements used in the telecom network. As cyber-attacks [36], [69] and physical attacks [35], [49] on technical infrastructure have become more widespread the probability of connection failure due to insufficient access controls is rated as MEDIUM .	Medium

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Implementation Flaw (Quality of the Product)	Availability (AP3-A)	Implementation flaws are considered as having MEDIUM probability to cause connection failure, as such flaws may have a variety of causes and may be non-trivial to detect, for example: - SW upgrades may result in the use of flawed or misconfigured SW [1], [66]. - Multi-vendor transport equipment or use of a mix of transmission technologies (incl. legacy technologies such as SONET/SDH) can lead to interoperability issues, especially in conjunction with SW updates. - Flawed dimensioning with respect to link capacity or route redundancy may cause network congestion. - Flawed architecture of the optical transport network may allow alien wavelength attacks [78] that degrade network performance.	Medium
Specification Weakness	Availability (AP4-A)	The probability of connection failure due to specification weaknesses in the 5G transport network is rated as LOW as the 5G transport network incorporates state-of-art transport protocols with robust QoS mechanisms defined by IETF.	Low
Failure of Power Supply	Availability (AP5-A)	Connection failure may be caused by a localized or regional power outage [42]. Such power outages can be caused by human error, power system failures, natural disasters, terrorist attacks, sabotage or theft of critical components, e.g. [49]. Given the diverse range of potential causes for power supply disruptions, the probability of these outages leading to connection failures is rated as MEDIUM .	Medium
Failure of Hardware Components or Unavailability of Spare Parts	Availability (AP6-A)	The probability of a hardware failure to cause connection failure is rated as MEDIUM , as there are a wide variety of components that may cause the failure, such as failed routing or switching equipment deployed at the radio sites in or data centers, cable breaks, physical damage of connectors, cabinets, etc., and a variety of failure sources such as human error, a natural disaster [8], a terror attack, e.g. [35], or sabotage or theft of components, e.g. [49]. HW failures were the single largest root cause of telecom security incidents in the EU in the year 2021 [38].	Medium

5.3.8 Tapping or Altering of the Geographical Position

Tapping or altering of geographical positions is not inherently tied to 5G backhaul and backbone components. These activities usually concern location-based services or GPS-related systems, which might be part of the broader 5G ecosystem but are not directly associated with the transport network infrastructure.

5.4 Private 5G Network Operator Perspective

The following additional assumptions (in addition to Chapter 5.1) apply for these risk scenarios for the private 5G network perspective.

- This scenario is only focused on on-premise networks.
- The network's backbone is part of the organization's general IT network.
- The network is separated by a firewall, which is part of the 5G network, from the public network.
- The network does not contain an IP Multimedia Subsystem (IMS).
- This scenario considers networks that are limited to dedicated locations.
- Organizations might operate multiple locations with separate private 5G networks.
- The members of the organization operating the network are also the user of the network. There is no separate group of users or a separate operator. [31]

This assessment assumes a private 5G network with typical network and security architectures as described in e.g. [82] and [83], respectively. Private networks are often deployed to improve the efficiency of the operations conducted on the campus, e.g. manufacturing facilities, logistics centers, mines, harbors, airports, railway yards, etc. Private networks may also be deployed at public sites such as hospitals, train stations, or exhibition centers.

Depending on the deployment, both RAN and core network including the transport network components may be hosted on the campus. Organizations may also operate multiple locations and in such case prefer the deployment of centralized core network functions and keep user plane traffic locally on campus. The network is typically separated by a firewall, which is part of the private campus network, from the public network, and the network's backbone is part of the organization's general IT network.

The transport network of a private 5G network may often be composed of heterogeneous network components provided by different network vendors and using different management systems:

- The IT network operated on the campus may often form the basis of the 5G transport network, e.g. Ethernet links may be used for backhauling between the core network and the 5G base stations, assuming the core network being hosted on the campus. In other deployments, transport equipment for backhaul and/or fronthaul may be selected by the RAN equipment vendor.
- The user plane of the 5G core network may use transport equipment selected by the equipment vendor of the 5G core network. This transport equipment, typically a service router, needs to be integrated with the IT network operated on the campus.
- The 5G user equipment may be attached to or possibly be integrated with transport equipment, e.g. an industrial field router.
- Operational Technology (OT) protocols operated over the private 5G network may require tunneling or protocol translation capabilities, possibly demanding for additional network elements.

The members of the organization operating the network may also be the users of the network. In this case, the network may be managed by operation teams having their primary skills in IT and/or OT domains. Alternatively, the network may be operated by a specialized operator, separate from the actual users of the network. In either case, the team(s) in charge of managing such a heterogeneous network deployment need to be knowledgeable with multiple transport technologies and transport equipment of multiple vendors.

For private wide area networks, e.g. for energy utilities, public safety, or the railways, the assessment of Section 5.2 largely applies, as private wide area networks have good communality with public MNO networks. A notable exception may be that the level of transport redundancy provided by a private wide area network can be far less as compared to a public MNO network, for example, redundancy of fiber optical transport may be limited as experienced in the attack on the private telecommunications network of Deutsche Bahn in October 2022 [49].

5.4.1 Network Failure

Network failures in 5G private on-premise networks refer to instances where the network infrastructure or components experience disruptions or malfunctions, resulting in service outages, connectivity issues, or degraded performance within the localized 5G network environment.

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Misconfiguration	Availability (AP1-A)	The transport network of a private 5G network is often composed of heterogeneous network components, both IT and OT components, provided by different network vendors and using different management systems [82], [83]. Misconfigurations such as flawed routing tables, port assignments or security settings may cause network outage, e.g. by disconnecting the core network from the RAN or the IT system of the organization. However, as those central parts of the transport network of a private 5G network do not change frequently and the network operations teams can be assumed as being knowledgeable with the most relevant IT technologies used in those parts of the transport network, such as TCP, UDP, IP or Ethernet [16], the probability of a misconfiguration causing network failure is rated as LOW .	Low
Insufficient Access Controls	Availability (AP2-A)	Physical access may enable an attacker to cause network failure, e.g. by disconnecting the core network from the RAN or the IT system of the organization. The transport network equipment connecting the core network is typically hosted in a well-protected and surveyed environment [17], [18]. The cables connecting the core network, particularly the cabling between the core network and the RAN, however, may be interrupted by attacking one or a small number of locations. While these locations are often well protected and surveyed, e.g. at manufacturing sites, deployment of the private network at public places such as stations or hospitals eases the physical access to these locations. This simplifies the destruction of the cabling between the core network and the RAN by targeted attacks or vandalism. Complete network outage may also be caused by cyber-attacks [36], [69] to the equipment or management system of the transport network, e.g. an intruder may misconfigure the router connecting the core network with the RAN and/or the IT system of the organization. From the above, the probability of insufficient access control to cause network failure by attacking the transport domain is rated as MEDIUM .	Medium

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Implementation Flaw (Quality of the Product)	Availability (AP3-A)	Implementation flaws are considered as having MEDIUM probability to cause network failure, as such flaws may have a variety of causes and may be non-trivial to detect, for example: - SW upgrades can lead to the use of flawed or misconfigured SW [1], [7], [66]. - Transport equipment provided by multiple vendors or use of a mix of transmission technologies including IT, OT, Ethernet, or passive optical LAN technologies, may cause interoperability issues, particularly in conjunction with SW updates. - Flawed dimensioning with respect to link capacity or route redundancy may cause network congestion.	Medium
Specification Weakness	Availability (AP4-A)	The probability of network failures due to specification weaknesses in the 5G transport network is rated as LOW as the 5G transport network incorporates state-of-art transport protocols with robust QoS mechanisms defined by IETF.	Low
Failure of Power Supply	Availability (AP5-A)	Failure of a private network may be caused by regional or localized power outage. Power outage may be caused by human error, power system failure, a natural disaster, a terror attack, e.g. [35], or sabotage or theft of components, e.g. [49]. Power outage may likely affect also other components of the private network or lead to a shut-down of the entire operation on the site (e.g. manufacturing site) on which network is deployed. Reducing the risk of a failure of power supply, a private network operator will likely use an uninterruptable power supply (UPS). Nevertheless, an UPS in a private 5G network is probably smaller in size in comparison to one in a public 5G network. Given the variety of different sources for failures of power supply, their probability to cause network failure is rated as HIGH .	High
Failure of Hardware Components or Unavailability of Spare Parts	Availability (AP6-A)	The probability of a hardware failure in the transport network to cause impairment or failure of the private network is rated as HIGH , as there are a wide variety of components that may cause the failure, such as failed routing or switching equipment, cable breaks, physical damage of connectors, etc., and a variety of failure sources such as human error, a natural disaster, a terror attack, e.g. [35], or sabotage or theft of components, e.g. [49]. Private networks may also be deployed without a good level of redundancy which may cause the entire network to fail if a hardware failure occurs at a centralized location in the network, for example, the router connecting the core network to the RAN or IT system of the organization.	High

5.4.2 Degraded Services / Connections

Degraded services / connections in a 5G private on-premise network signify a drop in quality or performance within the localized 5G environment. It may present as slower data transfer, increased latency, intermittent connectivity, packet loss, or reduced throughput compared to the expected levels. Factors such as signal interference, equipment issues, network congestion, or suboptimal configurations could contribute to this degradation. Considering the complexity and potential impact on operational efficiency, addressing the risk of degraded services/connections in on-premise 5G transport private networks involves a combination of comprehensive training, proactive monitoring, and swift incident response. Strengthening the capabilities of private network operators is essential for minimizing the impact of potential attacks on essential network components, ensuring the reliability and resilience of the on-premise network.

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Misconfiguration	Availability (AP1-A)	The transport network of a private 5G campus network is often composed of heterogeneous network components, both IT and OT components, provided by different network vendors and using different management systems [82], [83]. Misconfigurations such as flawed routing tables or port assignments, or misconfigurations related with quality of service, security or protocol translation parameters may degrade network performance, or cause degradation or outage to some services. Given the large number of network parameters and variety of services that need to be controlled, the probability of a misconfiguration causing degraded services / connections is rated as MEDIUM .	Medium
Insufficient Access Controls	Availability (AP2-A)	Physical access may enable an attacker to degrade services / connections, e.g. by disabling one or multiple aggregation routers deployed close to the radio sites or by cutting fronthaul or backhaul cables. The radio sites and backhaul or fronthaul cables of a private network are often well protected and surveyed, e.g. if deployed at manufacturing sites. Deployment at public places such as stations or hospitals eases the physical access to these sites and/or cables. This simplifies the destruction of backhaul or fronthaul cables or aggregation routers deployed at radio sites by targeted attacks or vandalism which in turn results in impaired network performance. From the above, the probability of insufficient access control in the transport network of a private 5G network to cause degraded services / connections is rated as MEDIUM .	Medium
Implementation Flaw (Quality of the Product)	Availability (AP3-A)	Implementation flaws are considered as having MEDIUM probability to cause degraded services / connections, as such flaws may have a variety of causes and may be non-trivial to detect, for example: • SW upgrades can lead to the use of flawed or misconfigured SW [1], [66]. • Transport equipment provided by multiple vendors or use of a mix of transmission technologies including IT, OT, Ethernet, or passive optical LAN technologies, may cause interoperability issues, particularly in conjunction with SW updates. • Flawed dimensioning with respect to link capacity or route redundancy may cause network congestion.	Medium
Specification Weakness	Availability (AP4-A)	The probability of degraded services / connections due to specification weaknesses in the 5G transport network is rated as LOW as the 5G transport network incorporates state-of-art transport protocols with robust QoS mechanisms defined by IETF.	Low
Failure of Power Supply	Availability (AP5-A)	Performance impairment of a private network, e.g. by disabling routing, switching or transmission equipment, may be caused by localized power outage. Power outage may be caused by human error, power system failure, a natural disaster, sabotage, etc. Reducing this risk, a private network operator will likely use an UPS. Nevertheless an UPS in a private 5G network is probably smaller in size in comparison to one in a public 5G network. Given the variety of different sources for localized failures of power supply, their probability to cause degraded services / connections is rated as HIGH .	High

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Failure of Hardware Components or Unavailability of Spare Parts	Availability (AP6-A)	The probability of a hardware failure to degrade services / connections is rated as HIGH , as there are a wide variety of components that may cause the failure, such as failed routing or switching equipment, cable breaks, physical damage of connectors, etc., and a variety of failure sources such as human error, a natural disaster, a terror attack, e.g. [35], or sabotage or theft of components, e.g. [49]. Private networks may also be deployed without a good level of redundancy which may cause degradations of network services if a hardware failure occurs at a centralized location in the network, for example, the router connecting the core network to the RAN or IT system of the organization.	High

5.4.3 Information Disclosure

Information disclosure in on-premise 5G transport private networks poses significant challenges, particularly due to the complexity and insider threat risks. Mitigating this risk requires a multi-faceted approach, encompassing access control, encryption practices, and comprehensive employee training. Protecting the confidentiality and privacy of data is paramount to maintaining the integrity and security of the network, safeguarding both organizational assets and customer trust.

Robust access control measures are essential to restrict and monitor access to sensitive information within the network. Encryption of sensitive data in transit and at rest is vital for safeguarding information from unauthorized access or exposure. Providing comprehensive security training to private network operators raises awareness about the importance of safeguarding sensitive information.

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Misconfiguration	Confidentiality (AP1-C)	Misconfigurations in network access settings can inadvertently expose sensitive data to unauthorized users. This can occur due to errors in router setting permissions, roles, or firewall rules. Misconfigured network protocols might not enforce encryption or secure communication channels, leading to potential data interception and exposure. Poorly configured security policies, such as inadequate password policies or unpatched software, increase the risk of unauthorized access and data leaks. In addition, diagnostic data, logs or other error reports may be transmitted over the network, containing sensitive information, e.g. IP addresses or configuration details, which can be misused. The probability of confidentiality issues with information disclosure in on-premise 5G transport private network due to misconfigurations is therefore rated as MEDIUM .	Medium
Insufficient Access Controls	Confidentiality (AP2-C)	Lack of robust authentication methods, such as multi-factor authentication (MFA), can allow unauthorized access to sensitive information. Insufficient access control can result in users having more privileges than necessary, increasing the risk of accidental or malicious data disclosure. Without continuous monitoring and auditing, unauthorized access to sensitive data can go unnoticed. For example, if an attacker gains access through compromised credentials, they can explore and exfiltrate data without being detected. Furthermore, physical access to routers and switches might allow attackers to perform side-channel attacks to extract sensitive information from the network. Therefore, the probability of confidentiality issues with information disclosure in on-premise 5G transport private network due insufficient access controls is rated as MEDIUM .	Medium

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Implementation Flaw (Quality of the Product)	Confidentiality (AP3-C)	Software vulnerabilities in network devices can be exploited by attackers. For example, a firmware flaw in a router can allow authentication to be bypassed and sensitive data to be accessed. A private 5G operator using routers from a particular vendor could face a firmware flaw that allows attackers to exploit a buffer overflow vulnerability, bypass authentication, access sensitive configurations, intercept traffic and manipulate settings to exfiltrate information. [56]. Therefore, the probability of confidentiality issues with information disclosure in on-premise private 5G transport networks due to implementation flaws is rated as MEDIUM .	Medium
Specification Weakness	Confidentiality (AP4-C)	If the network design specifications do not include comprehensive security requirements, critical vulnerabilities may go unnoticed. For example, specifications that do not mandate encryption for all data transmissions can lead to data exposure. Weaknesses in the testing phase, where security aspects are not thoroughly tested, can result in undetected vulnerabilities being present in the deployed network. Using outdated or insecure protocols as specified in the network design can leave data transmissions vulnerable to interception and unauthorized access. Nevertheless, specifications are regularly adapted to state-of-the-art cryptography standards, resulting in a LOW probability for information disclosure scenarios due to specification weaknesses.	Low
Failure of Power Supply	Confidentiality (AP5-C)	Failure of power supply appears unrelated to information disclosure.	Low
Failure of Hardware Components or Unavailability of Spare Parts	Confidentiality (AP6-C)	If hardware failure leads to rerouting of traffic through unsecured or compromised routing paths, attackers might exploit the rerouted traffic and perform man-in-the-middle attacks on the vulnerable routing path. However, this scenario has a LOW probability to occur.	Low

5.4.4 User Identities (Authenticity/Integrity)

Given the context of an on-premise 5G network scenario, tapping or misusing user identities is not a relevant risk scenario within the 5G transport network.

6 Risk Assessment Results

This chapter presents the final results of this detailed risk analysis. The final risk analysis that covers all subsystems is presented in the framework document.

Note: Risk scenarios with no given probability level were not rated, see beginning of Chapter 5 and the corresponding risk scenario.

6.1 Probability Assessment

The following table summarizes the results of this detailed analysis of the **transport network**. According to the maximum principle in risk analysis only the highest probability is listed.

Perspective	Section	Risk Scenario	Probability Level
Public MNO	5.2.1	Emergency Call (TKG § 164)	Medium
	5.2.2	Cell Broadcast (TKG § 164a)	Medium
	5.2.3	Telecommunications Secrecy (TTDSG § 3)	Medium
	5.2.4	Emergency Preparedness (TKG § 184)	Medium
	5.2.5	Regional Network Failure	Medium
	5.2.6	Multi-Regional Network Failure	Medium
	5.2.7	Failure of Mobile Service	Medium
	5.2.8	Fraud	
	5.2.9	Subscriber-Data Leak	Medium
User	5.3.1	Urgent Telephone Calls Not Possible	Medium
	5.3.2	Authenticity of the Receiver of a Call	
	5.3.3	Integrity of Transmitted Data	Medium
	5.3.4	User's Identity	
	5.3.5	Eavesdropping of a Communication	Medium
	5.3.6	Slowing Down/Restricting the Connection	Medium
	5.3.7	Connection Failure	Medium
	5.3.8	Tapping or Altering of the Geographical Position	
Private 5G Network Operator	5.4.1	Network Failure	High
	5.4.2	Degraded Services/Connections	High
	5.4.3	Information Disclosure	Medium
	5.4.4	User Identities (Authenticity/ Integrity)	

6.2 Risk Assessment

The following table summarizes the overall risk assessment of the **transport network**. The risk scenarios and their respective impact level are described in the framework document.

Perspective	Risk Scenario	Impact	Probability	Risk Score
Public MNO	Emergency Call (TKG § 164)	High	Medium	High
	Cell Broadcast (TKG § 164a)	High	Medium	High
	Telecommunications Secrecy (TTDSG § 3)	Medium	Medium	Medium
	Emergency Preparedness (TKG § 184)	High	Medium	High
	Regional Network Failure	Medium	Medium	Medium
	Multi-Regional Network Failure	High	Medium	High
	Failure of Mobile Service	Medium	Medium	Medium
	Fraud			
	Subscriber-Data Leak	Medium	Medium	Medium
User	Urgent Telephone Calls Not Possible	Low	Medium	Low
	Authenticity of the Receiver of a Call			
	Integrity of Transmitted Data	High	Medium	High
	User's Identity			
	Eavesdropping of a Communication	Medium	Medium	Medium
	Slowing Down/Restricting the Connection	Low	Medium	Low
	Connection Failure	Low	Medium	Low
	Tapping or Altering of the Geographical Position			
Private 5G Network Operator	Network Failure	Medium	High	High
	Degraded Services/Connections	Medium	High	High
	Information Disclosure	Medium	Medium	Medium
	User Identities (Authenticity/ Integrity)			

Bibliography

- [1] Antrobus, B., "Optus reveals cause of massive network outage which crippled nation," News.Com.au, November 2023
<https://www.news.com.au/technology/online/security/optus-reveals-cause-of-massive-network-outage-which-crippled-nation/news-story/02f5b50835c6a01586c5bd880034c0e0>
- [2] The Sun UK Edition, "THOUSANDS of Three users were left without a phone signal as the network suffers an outage," The Sun UK Edition, December 2023
<https://www.thesun.co.uk/money/24914467/three-down-mobile-network-service-status-updates>
- [3] REUTERS Media & Telecom 2022, "Europe braces for mobile network blackouts," Reuters, September 2022
<https://www.reuters.com/business/media-telecom/exclusive-europe-braces-mobile-network-blackouts-sources-2022-09-29>
- [4] The GUARDIAN, "Telstra downplays possibility of allowing customers to roam rival networks during outages," The GUARDIAN, November 2023
<https://www.theguardian.com/business/2023/nov/22/telstra-roaming-customers-roam-rival-networks-optus-outage-inquiry>
- [5] The Asahi Shimbun National Report, "KDDI users still affected by major network failure two days later," The Asahi Shimbun National Report, July 2022
<https://www.asahi.com/ajw/articles/14660642>
- [6] The Japantimes, "KDDI's mobile phone network restored after outage across Japan," The Japantimes, December 2023
<https://www.japantimes.co.jp/business/2023/12/11/companies/kddi-nationwide-outage/>
- [7] BBC, "Canada hit by massive mobile and internet outage," BBC, July 2022
<https://www.bbc.com/news/world-us-canada-62102223>
- [8] CBC News, "Bell CEO says company working toward speedier repairs after powerful storms," CBC News, October 2023
<https://www.cbc.ca/news/canada/nova-scotia/bell-speedier-repairs-after-extreme-weather-events-1.6986872>
- [9] Lipscombe, P., "T-Mobile suffers network outage with customers disrupted across US," DCD News, February 2023
<https://www.datacenterdynamics.com/en/news/t-mobile-suffers-network-outage-with-customers-disrupted-across-us/>
- [10] Tagesschau News, "Hacker attack paralyses Ukrainian mobile phone providers," Tagesschau, December 2023
<https://www.tagesschau.de/ausland/europa/ukraine-hacker-mobilfunk-russland-100.html>
- [11] TECHJUICE News, "Major Internet Service Providers Goes Down Causing Connectivity Issues Across Pakistan," TECHJUICE, August 2022
<https://www.techjuice.pk/internet-services-are-down-all-over-pakistan/>
- [12] Belson, D., "Internet disruptions overview for Q1 2023," The Cloudflare Blog, December 2023
<https://blog.cloudflare.com/q1-2023-internet-disruption-summary/>
- [13] Belson, D., "Internet disruptions overview for Q2 2022," The Cloudflare Blog, May 2022
<https://blog.cloudflare.com/q2-2022-internet-disruption-summary/>
- [14] Infinera Corporation, "Synchronization Distribution in 5G Transport Networks," Infinera E-Book, 2021
<https://docslib.org/doc/6944097/e-book-synchronization-distribution-in-5g-transport-networks>

- [15] NSA, "Network Slicing: Security Considerations for Design, Deployment, and Maintenance," NSA Media Relations and CISA Media Relations, July 2023
<https://media.defense.gov/2023/Jul/17/2003260829/-1/-1/0/ESF%205G%20NETWORK%20SLICING-SECURITY%20CONSIDERATIONS%20FOR%20DESIGN,%20DEPLOYMENT,%20AND%20MAINTENANCE%20FINAL.PDF>
- [16] Huawei, "5G Transport Network," Huawei IP Encyclopedia, April 2023
<https://info.support.huawei.com/info-finder/encyclopedia/en/5G+Transport+Network.html>
- [17] Boyle, K., "5G is all in the timing," Ericsson Blog, August 2019
<https://www.ericsson.com/en/blog/2019/8/what-you-need-to-know-about-timing-and-sync-in-5g-transport-networks>
- [18] Ericsson White Paper, "Enabling time-critical applications over 5G with rate adaptation," Ericsson, May 2021
<https://www.ericsson.com/en/reports-and-papers/white-papers/enabling-time-critical-applications-over-5g-with-rate-adaptation/>
- [19] ENISA, "ENISA Threat Landscape for 5G Networks Report," December 2020
<https://www.enisa.europa.eu/publications/enisa-threat-landscape-report-for-5g-networks>
- [20] (Protocols) MDPI Journal 2023, Networking Architectures and Protocols, MDPI, May 2023
<https://www.mdpi.com/2079-9292/12/11/2490>
- [21] Mahan, J., "Environmental Monitoring: Catch Data Center Equipment Failure," C&C Technology Group, August, 2022
<https://cc-techgroup.com/environmental-monitoring>
- [22] Alavikia, Z. and Shabro, M., "A comprehensive layered approach for implementing internet of things-enabled smart grid: A survey," ScienceDirect Journals & Books, June 2022
<https://www.sciencedirect.com/science/article/pii/S2352864822000025>
- [23] Saunders, J., "The vital role of microwave and millimeter wave for backhauling 5G traffic (Analyst Angle)," RCRWireless News, January 2021
<https://www.rcrwireless.com/20210108/5g/the-vital-role-of-microwave-and-millimeter-wave-for-backhauling-5g-traffic-analyst-angle>
- [24] Park, S., et.al., "5G Security Threat Assessment in Real Networks," MDPI, August 2021
<https://www.mdpi.com/1424-8220/21/16/5524>
- [25] ENISA, "5G SUPPLEMENT," The European Union Agency for Cybersecurity (ENISA), July 2021
<https://www.enisa.europa.eu/publications/5g-supplement-security-measures-under-eccc/@download/fullReport>
- [26] Ericsson Microwave Outlook, "When long haul makes a difference," 10th Edition, October 2023
<https://www.ericsson.com/en/reports-and-papers/microwave-outlook/articles/when-long-haul-makes-a-difference>
- [27] ECC Decision (20)01, "On the harmonised use of the frequency band 5945-6425 MHz for Wireless Access Systems including Radio Local Area Networks (WAS/RLAN)," November 2020
[https://docdb.cept.org/download/50365191-a99d/ECC%20Decision%20\(20\)01.pdf](https://docdb.cept.org/download/50365191-a99d/ECC%20Decision%20(20)01.pdf)
- [28] NIST SPECIAL PUBLICATION 1800-33B, "5G Cybersecurity", April 2022
<https://www.nccoe.nist.gov/sites/default/files/2022-04/nist-5G-sp1800-33b-preliminary-draft.pdf>
- [29] CQR, "Insufficient Cryptographic Protection of Data in Transit," CQR Blog, February 2023
<https://cqr.company/web-vulnerabilities/insufficient-cryptographic-protection-of-data-in-transit/>

- [30] Saurugg, H., "Europe on the road to a major disaster!?", Gesellschaft für KRISENVORSORGE, April 2022
<https://www.saurugg.net/wp-content/uploads/2022/05/europe-on-the-road-to-disaster-update-04-2022.pdf>
- [31] National Institute of Standards and Technology U.S. Department of Commerce, "Guidelines for Managing the Security of Mobile Devices in the Enterprise," NIST, May 2023
<https://csrc.nist.gov/pubs/sp/800/124/r2/ipd>
- [32] Frost & Sullivan, "Road to 5G: IP Transport Network is the top priority for Telecom Operators," Frost & Sullivan, December 2019
<https://www.frost.com/frost-perspectives/road-to-5g-ip-transport-network-is-the-top-priority-for-telecom-operators/>
- [33] Cisco's official Security Advisory page 2022, "Cisco IOS XE Software IPv6 VPN over MPLS Denial of Service Vulnerability," Cisco's official Security Advisory page, September 2022
<https://sec.cloudapps.cisco.com/security/center/content/CiscoSecurityAdvisory/cisco-sa-iosxe-6vpe-dos-tJbt5Zv>
- [34] ETSI, "IPv6 Best Practices Benefits Transition Challenges and the Way Forward," ETSI White Paper, August 2020
https://www.etsi.org/images/files/ETSIWhitePapers/etsi_WP35_IPv6_Best_Practices_Benefits_Transition_Challenges_and_the_Way_Forward
- [35] Deutsche Welle, "Germany: Tesla plant targeted in 'deliberate arson' — police," March 2024
<https://www.dw.com/en/germany-tesla-plant-targeted-in-deliberate-arson-police/a-68455762>
- [36] Brooks, K. J., "What customers should know about AT&T's massive data breach," CBS News, April 2024
<https://www.cbsnews.com/news/att-data-breach-2024-cbs-news-explains>
- [37] Securelist, Router security in 2021, Kaspersky, June 2022
<https://securelist.com/router-security-2021/106711/>
- [38] ENISA, "Telecom Security Incidents 2021," ENISA Annual Report, July 2022
<https://www.enisa.europa.eu/publications/telecom-security-incidents-2021>
- [39] Nokia 1830 Security Management Server, Data Sheet, 2023
https://onestore.nokia.com/asset/200458?_gl=1*uz3byh*_gcl_au*MTA2MDA0MDQ0Mi4xNzE5MjIyMjQ4*_ga*MTcwODkwNTE3OC4xNzE5MjIyMTU3*_ga_D6GE5QF247*MTcxOTI0MzQ4Mi40LjEuMTcxOTI0MzUxOS4wLjAuMA..&_ga=2.93133022.2110879277.1719222157-1708905178.1719222157
- [40] America's Cyber Defense Agency, "NSA Releases Guidance on Eliminating Obsolete TLS Protocol Configurations," CISA, January 2021
<https://www.cisa.gov/news-events/alerts/2021/01/05/nsa-releases-guidance-eliminating-obsolete-tls-protocol>
- [41] Barker, E., "Recommendation for Key Management: Part 1 - General," NIST, May 2020
<https://csrc.nist.gov/pubs/sp/800/57/pt1/r5/final>
- [42] Prince, M., et.al., "Major data center power failure (again): Cloudflare Code Orange tested," Cloudflare Blog, April 2024
<https://www.sciencedirect.com/science/article/pii/S1573427721000163>
- [43] The Hacker News, "Why Human Error is #1 Cyber Security Threat to Businesses in 2021," February 2021
<https://thehackernews.com/2021/02/why-human-error-is-1-cyber-security.html>
- [44] FitzGerald, D., and McMillan, R., "T-Mobile hacker who stole data on 50 million customers: 'Their security is awful'," August 2021
<https://www.foxbusiness.com/technology/t-mobile-hacker-stole-data-50-million-customers-security-awful>

- [45] Young, T., "GTP Remains a 5G Security Threat as Operators Transition to 5G," A10 Networks, December 2019
<https://www.a10networks.com/blog/gtp-remains-a-5g-security-threat-as-operators-transition-to-5g/>
- [46] Jones, C., "Some 300,000 Ips vulnerable to this Loop DoS attack," The Register, March 2024
https://www.theregister.com/2024/03/24/loop_ip_vulnerable/
- [47] Clague, D., "Fiber Tapping and Data Security: Unraveling the Potential Threats and Detection Methods," Viavi, December 2023
<https://blog.viavisolutions.com/2023/12/06/fiber-tapping-and-data-security-unraveling-the-potential-threats-and-detection-methods/>
- [48] Kedrosky, E., "Real Life Examples of AWS and Azure Privilege Escalation," Sonrai Security, August 2022
<https://sonraisecurity.com/blog/real-life-examples-of-privilege-escalation-in-aws-and-azure/>
- [49] Wikipedia, "October 2022 German railway attack," October 2022
https://en.wikipedia.org/wiki/October_2022_German_railway_attack
- [50] Network World, "5G network slices could be vulnerable to attack, researchers say," May 2023
<https://www.networkworld.com/article/972286/5g-network-slices-could-be-vulnerable-to-attack-researchers-say.html>
- [51] Seals, T., "Black Hat 2019: 5G Security Flaw Allows MiTM, Targeted Attacks," August 2019
<https://threatpost.com/5g-security-flaw-mitm-targeted-attacks/147073/>
- [52] Fonyi, S., "Cyber Defense Army, "Overview of 5G Security and Vulnerabilities," March 2020
https://cyberdefensereview.army.mil/Portals/6/CDR%20V5N1%20-%2008_%20Fonyi_WEB.pdf
- [53] Dzhekishev, B., "Big Data Security Best Practices," MAD DEVS, September 2023
<https://maddevs.io/blog/big-data-security-best-practices/>
- [54] Higgins, K.J., "Unpatched Vulnerabilities the Source of Most Data Breaches," Dark Reading, April 2018
<https://www.darkreading.com/vulnerabilities-threats/unpatched-vulnerabilities-the-source-of-most-data-breaches>
- [55] Brandom, R., "Almost all WannaCry victims were running Windows 7", The Verge, May 2017
<https://www.theverge.com/2017/5/19/15665488/wannacry-windows-7-version-xp-patched-victim-statistics>
- [56] 5G Americas White Paper, "Transport networks for 5G," August 2023
<https://www.5gamericas.org/wp-content/uploads/2023/07/Transport-Networks-for-5G-InDesign.pdf>
- [57] Foo, M., "The use of E-Band for backhaul, a key ingredient for successful 5G & beyond", ABI Research, August 2023
<https://www.rcrwireless.com/20230807/5g/the-use-of-e-band-for-backhaul-a-key-ingredient-for-successful-5g-beyond-analyst-angle>
- [58] Nokia Case Study, "E-Band microwave backhaul performance field-proven under heavy rainfall," 2023
https://onestore.nokia.com/asset/213320?_ga=2.108099394.1737266075.1713529940-1021459066.1713529940
- [59] Gautam, A., "8 Biggest Cloud Security Breaches You Should Know About," BuildSomeTech, July 2022
<https://www.buildsometechnology.com/biggest-cloud-security-breaches/>
- [60] Giannotti, L., "The biggest cyberattacks of 2024," TechMonitor30, May 2024
<https://techmonitor.ai/technology/cybersecurity/biggest-cyberattacks-of-2024>
- [61] Dugan, J. et. Al, "Social vulnerability to long-duration power outages," International Journal of Disaster Risk Reduction, Vol. 85, February 2023
<https://www.sciencedirect.com/science/article/pii/S2212420922007208>

- [62] Ericsson, "A guide to 5G network security 2.0: Conceptualizing security in mobile communication networks – how does 5G fit in?," September 2021
<https://www.ericsson.com/4a66f8/assets/local/security/09172021-a-guide-to-5g-network-security-2.0.pdf>
- [63] Marget, A, "Downtime: Causes, Costs and How to Minimize It," Unitrends
<https://www.unitrends.com/blog/downtime-causes-costs-and-how-to-minimize-it>
- [64] Moss, S., "Cloudflare recovers from service outage after power failure at core North American data center," Data Centre Dynamics, November 2023
<https://www.datacenterdynamics.com/en/news/cloudflare-recovers-from-service-outage-after-power-failure-at-core-north-american-data-center/>
- [65] IETF, RFC5920, "Security Framework for MPLS and GMPLS Networks", July 2010
<https://datatracker.ietf.org/doc/rfc5920/>
- [66] Zahn, M., et.al., , "AT&T outage caused by software update, company says", ABC News, February 2024
<https://abcnews.go.com/US/att-outage-impacting-us-customers-company/story?id=107440297>
- [67] The Hacker News, "Mysterious Cyber Attack Took Down 600,000+ Routers in the U.S.," May 2024
<https://thehackernews.com/2024/05/mysterious-cyber-attack-takes-down.html>
- [68] FCC, "A Report of the Public Safety and Homeland Security Bureau Federal Communications Commission," October 2020
<https://docs.fcc.gov/public/attachments/DOC-367699A1.pdf>
- [69] Greig, J., "Vodafone Portugal hit with cyberattack affecting 4G/5G network, TV, SMS services," ZDNET, February 2022
<https://www.zdnet.com/article/vodafone-portugal-hit-with-cyberattack-affecting-4g5g-network-tv-sms-services-and-more/>
- [70] Encryption Consulting, "Why older TLS protocols are unsafe for your organization?," May 2024
<https://www.encryptionconsulting.com/why-should-organizations-avoid-older-tls-protocols/#:~:text=Over%20time%2C%20new%20TLS%20versions,TLS%201.0%2C%20and%20TLS%201.1.>
- [71] Madory, D., "A Brief History of the Internet's Biggest BGP Incidents From Traffic-Disrupting BGP Leaks to Crypto-Stealing BGP Hijacks", NANOG, 5 September 2023
<https://www.nanog.org/stories/articles/a-brief-history-of-the-internets-biggest-bgp-incidents/>
- [72] Casey, M., "Firewall to deter cyberattacks is blamed for Massachusetts 911 outage," June 2024
<https://apnews.com/article/911-outage-massachusetts-79b48c44a587da8a91f16f4758cd88ab>
- [73] White, M., "Dutch emergency services hit by major KPN telecoms outage," June 2019
<https://www.standard.co.uk/news/world/dutch-emergency-services-hit-by-major-kpn-telecoms-outage-a4175011.html>
- [74] Whittaker, Z., "Amazon explains latest cloud outage: Blame the power," ZDNET, June 2012
<https://www.zdnet.com/article/amazon-explains-latest-cloud-outage-blame-the-power/>
- [75] Euronews, "French telecom operator Orange apologises after emergency numbers crash nationwide," June 2021
<https://www.euronews.com/2021/06/03/french-telecom-operator-orange-apologises-after-emergency-numbers-crash-nationwide>
- [76] Jim, Z. and Elber, J., "Optical Transport for 5G Mobile Network: Challenges and Solutions," IEEE Future Networks, October 2021
<https://futurenetworks.ieee.org/tech-focus/october-2021/optical-transport-for-5g-mobile-network-challenges-and-solutions>

- [77] IEA, "Electricity security matters more than ever," October 2020
https://iea.blob.core.windows.net/assets/cd69028a-da78-4b47-b1bf-7520cdb20d70/Power_systems_in_transition.pdf
- [78] Furdek, M., et al., "Vulnerabilities and Security Issues in Optical Networks," July 2014
https://www.researchgate.net/publication/269268194_Vulnerabilities_and_security_issues_in_optical_networks
- [79] ETSI TR 102 445 V1.2.1, "Emergency Communications (EMTEL); Overview of Emergency Communications Network Resilience and Preparedness," April 2023
https://www.etsi.org/deliver/etsi_tr/102400_102499/102445/01.02.01_60/tr_102445v010201p.pdf
- [80] Pollina, E., et al., "Emergency call services, telcos urge EU to protect telecoms networks from power cuts," Reuters, December 2022
<https://www.reuters.com/business/media-telecom/emergency-call-services-telcos-urge-eu-protect-telecoms-networks-power-cuts-2022-12-05/>
- [81] Moriarty, K. and Farrell, S., "RFC 8996 Deprecating TLS 1.0 and TLS 1.1," RFC Editor, March 2021
<https://www.rfc-editor.org/rfc/rfc8996>
- [82] 5G-ACIA, "Industrial 5G Edge Computing – Use Cases, Architecture and Deployment," White Paper, February 2023
<https://5g-acia.org/whitepapers/industrial-5g-edge-computing-use-cases-architecture-and-deployment/>
- [83] 5G-ACIA, "Security Aspects of 5G for Industrial Networks," White Paper, February 2021
<https://5g-acia.org/whitepapers/security-aspects-of-5g-for-industrial-networks/>
- [84] IETF RFC 6071, "IP Security (IPsec) and Internet Key Exchange (IKE) Document Roadmap," February 2011
<https://datatracker.ietf.org/doc/html/rfc6071>
- [85] IETF RFC 8446, "The Transport Layer Security (TLS) Protocol Version 1.3," August 2018
<https://datatracker.ietf.org/doc/html/rfc8446>
- [86] Welekwe, A., "Buffer overflow vulnerabilities and attacks explained," Comparitech, September 2023
<https://www.comparitech.com/blog/information-security/buffer-overflow-attacks-vulnerabilities/>
- [87] Ciphersuite.Info, Weak Cipher Suites
<https://ciphersuite.info/cs/?security=weak>
- [88] America's Cyber Defence Agency, Risks of Default Passwords on the Internet, October 2016
<https://www.cisa.gov/news-events/alerts/2013/06/24/risks-default-passwords-internet>
- [89] comparitech, One in 16 home wi-fi routers tested vulnerable to default password attacks: report, August 2023
<https://www.comparitech.com/blog/information-security/default-password-routers-study/>
- [90] Bays et al., Virtual network security: threats, countermeasures, and challenges, January 2015
<https://jisajournal.springeropen.com/articles/10.1186/s13174-014-0015-z>
- [91] Emily Thomas, Supply chain issues delay Wi-Fi router upgrades, October 2022
<https://calvinchimes.org/2022/10/02/supply-chain-issues-delay-wi-fi-router-upgrades/>
- [92] TeleTraders, Impact of Supply Chain Issues when Upgrading Your Network, December 2023
<https://teletraders.net/impact-of-supply-chain-issues-when-upgrading-your-network/>
- [93] Kaspersky encyclopedia, Downgrade attack
<https://encyclopedia.kaspersky.com/glossary/downgrade-attack/>

- [94] Praetorian, Man-in-the-Middle TLS Protocol Downgrade Attack, August 2014
<https://www.praetorian.com/blog/man-in-the-middle-tls-ssl-protocol-downgrade-attack/>